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1. Replay in Deep Learning: Current Approaches and Missing Biological Elements

Tyler L. Hayes, Giri P. Krishnan, Maxim Bazhenov, Hava T. Siegelmann, Terrence J. Sejnowski, Christopher Kanan

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Replay is the reactivation of one or more neural patterns that are similar to the activation patterns experienced during past waking experiences. Replay was first observed in biological neural networks during sleep, and it is now thought to play a critical role in memory formation, retrieval, and consolidation. Replay-like mechanisms have been incorporated in deep artificial neural networks that learn over time to avoid catastrophic forgetting of previous knowledge. Replay algorithms have been successfully used in a wide range of deep learning methods within supervised, unsupervised, and reinforcement learning paradigms. In this letter, we provide the first comprehensive comparison between replay in the mammalian brain and replay in artificial neural networks. We identify multiple aspects of biological replay that are missing in deep learning systems and hypothesize how they could be used to improve artificial neural networks.

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2. Parametric UMAP Embeddings for Representation and Semisupervised Learning

Tim Sainburg, Leland McInnes, Timothy Q. Gentner

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UMAP is a nonparametric graph-based dimensionality reduction algorithm using applied Riemannian geometry and algebraic topology to find low-dimensional embeddings of structured data. The UMAP algorithm consists of two steps: (1) computing a graphical representation of a data set (fuzzy simplicial complex) and (2) through stochastic gradient descent, optimizing a low-dimensional embedding of the graph. Here, we extend the second step of UMAP to a parametric optimization over neural network weights, learning a parametric relationship between data and embedding. We first demonstrate that parametric UMAP performs comparably to its nonparametric counterpart while conferring the benefit of a learned parametric mapping (e.g., fast online embeddings for new data). We then explore UMAP as a regularization, constraining the latent distribution of autoencoders, parametrically varying global structure preservation, and improving classifier accuracy for semisupervised learning by capturing structure in unlabeled data.1.

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3. Dynamic Spatiotemporal Pattern Recognition with Recurrent Spiking Neural Network

Jiangrong Shen, Jian K. Liu, Yueming Wang

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Our real-time actions in everyday life reflect a range of spatiotemporal dynamic brain activity patterns, the consequence of neuronal computation with spikes in the brain. Most existing models with spiking neurons aim at solving static pattern recognition tasks such as image classification. Compared with static features, spatiotemporal patterns are more complex due to their dynamics in both space and time domains. Spatiotemporal pattern recognition based on learning algorithms with spiking neurons therefore remains challenging. We propose an end-to-end recurrent spiking neural network model trained with an algorithm based on spike latency and temporal difference backpropagation. Our model is a cascaded network with three layers of spiking neurons where the input and output layers are the encoder and decoder, respectively. In the hidden layer, the recurrently connected neurons with transmission delays carry out high-dimensional computation to incorporate the spatiotemporal dynamics of the inputs. The test results based on the data sets of spiking activities of the retinal neurons show that the proposed framework can recognize dynamic spatiotemporal patterns much better than using spike counts. Moreover, for 3D trajectories of a human action data set, the proposed framework achieves a test accuracy of 83.6% on average. Rapid recognition is achieved through the learning methodology-based on spike latency and the decoding process using the first spike of the output neurons. Taken together, these results highlight a new model to extract information from activity patterns of neural computation in the brain and provide a novel approach for spike-based neuromorphic computing.

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4. Temporal Variabilities Provide Additional Category-Related Information in Object Category Decoding: A Systematic Comparison of Informative EEG Features

Hamid Karimi-Rouzbahani, Mozhgan Shahmohammadi, Ehsan Vahab, Saeed Setayeshi, Thomas A. Carlson

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How does the human brain encode visual object categories? Our understanding of this has advanced substantially with the development of multivariate decoding analyses. However, conventional electroencephalography (EEG) decoding predominantly uses the mean neural activation within the analysis window to extract category information. Such temporal averaging overlooks the within-trial neural variability that is suggested to provide an additional channel for the encoding of information about the complexity and uncertainty of the sensory input. The richness of temporal variabilities, however, has not been systematically compared with the conventional mean activity. Here we compare the information content of 31 variability-sensitive features against the mean of activity, using three independent highly varied data sets. In whole-trial decoding, the classical event-related potential (ERP) components of P2a and P2b provided information comparable to those provided by original magnitude data (OMD) and wavelet coefficients (WC), the two most informative variability-sensitive features. In time-resolved decoding, the OMD and WC outperformed all the other features (including the mean), which were sensitive to limited and specific aspects of temporal variabilities, such as their phase or frequency. The information was more pronounced in the theta frequency band, previously suggested to support feedforward visual processing. We concluded that the brain might encode the information in multiple aspects of neural variabilities simultaneously such as phase, amplitude, and frequency rather than mean per se. In our active categorization data set, we found that more effective decoding of the neural codes corresponds to better prediction of behavioral performance. Therefore, the incorporation of temporal variabilities in time-resolved decoding can provide additional category information and improved prediction of behavior.

5. Flexible Transmitter Network

Shao-Qun Zhang, Zhi■Hua Zhou

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Current neural networks are mostly built on the MP model, which usually formulates the neuron as executing an activation function on the real-valued weighted aggregation of signals received from other neurons. This letter proposes the flexible transmitter (FT) model, a novel bio-plausible neuron model with flexible synaptic plasticity. The FT model employs a pair of parameters to model the neurotransmitters between neurons and puts up a neuron-exclusive variable to record the regulated neurotrophin density. Thus, the FT model can be formulated as a two-variable, two-valued function, taking the commonly used MP neuron model as its particular case. This modeling manner makes the FT model biologically more realistic and capable of handling complicated data, even spatiotemporal data. To exhibit its power and potential, we present the flexible transmitter network (FTNet), which is built on the most common fully connected feedforward architecture taking the FT model as the basic building block. FTNet allows gradient calculation and can be implemented by an improved backpropagation algorithm in the complex-valued domain. Experiments on a broad range of tasks show that FTNet has power and potential in processing spatiotemporal data. This study provides an alternative basic building block in neural networks and exhibits the feasibility of developing artificial neural networks with neuronal plasticity.

6. Neural Networks with Disabilities: An Introduction to Complementary Artificial Intelligence

Vagan Terziyan, Olena Kaikova

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Machine learning is a good tool to simulate human cognitive skills as it is about mapping perceived information to various labels or action choices, aiming at optimal behavior policies for a human or an artificial agent operating in the environment. Regarding autonomous systems, objects and situations are perceived by some receptors as divided between sensors. Reactions to the input (e.g., actions) are distributed among the particular capability providers or actuators. Cognitive models can be trained as, for example, neural networks. We suggest training such models for cases of potential disabilities. Disability can be either the absence of one or more cognitive sensors or actuators at different levels of cognitive model. We adapt several neural network architectures to simulate various cognitive disabilities. The idea has been triggered by the "coolability" (enhanced capability) paradox, according to which a person with some disability can be more efficient in using other capabilities. Therefore, an autonomous system (human or artificial) pretrained with simulated disabilities will be more efficient when acting in adversarial conditions. We consider these coolabilities as complementary artificial intelligence and argue on the usefulness if this concept for various applications.

7. Asymptotic Input-Output Relationship Predicts Electric Field Effect on Sublinear Dendritic Integration of AMPA Synapses

Yaqin Fan, Xile Wei, Guosheng Yi, Meili Lu, Jiang Wang, Bin Deng

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An extracellular electric field (EF) induces transmembrane polarizations on extremely inhomogeneous spaces. Evidence shows that EF-induced somatic polarization in pyramidal cells can modulate the neuronal input-output (I/O) function. However, it remains unclear whether and how dendritic polarization participates in the dendritic integration and contributes to the neuronal I/O function. To this end, we built a computational model of a simplified pyramidal cell with multi-dendritic tufts, one dendritic trunk, and one soma to describe the interactions among EF, dendritic integration, and somatic output, in which the EFs were modeled by inserting inhomogeneous extracellular potentials. We aimed to establish the underlying relationship between dendritic polarization and dendritic integration by analyzing the dynamics of subthreshold membrane potentials in response to AMPA synapses in the presence of constant EFs. The model-based singular perturbation analysis showed that the equilibrium mapping of a fast subsystem can serve as the asymptotic subthreshold I/O relationship for sublinear dendritic integration. This allows us to predict the tendency of EF-mediated dendritic integration by showing how EF changes modify equilibrium mapping. EF-induced hyperpolarization of distal dendrites receiving synapses inputs was found to play a key role in facilitating the AMPA receptor-evoked excitatory postsynaptic potential (EPSP) by enhancing the driving force of synaptic inputs. A significantly higher efficacy of EF modulation effect on global AMPA-type dendritic integration was found compared with local AMPA-type dendritic integration. During the generation of an action potential (AP), the relative contribution of EF-modulated dendritic integration and EF-induced somatic polarization was determined to show their collaboration in promoting or inhibiting the somatic excitability, depending on the EF polarity. These findings are crucial for understanding the EF modulation effect on neuronal computation, which provides insight into the modulation mechanism of noninvasive brain modulation.

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8. Expansion of Information in the Binary Autoencoder with Random Binary Weights

Viacheslav Osaulenko

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This letter studies the expansion and preservation of information in a binary autoencoder where the hidden layer is larger than the input. Such expansion is widespread in biological neural networks, as in the olfactory system of a fruit fly or the projection of thalamic inputs to the neocortex. We analyze the threshold model, the kWTA model, and the binary matching pursuit model to find how the sparsity and the dimension of the encoding influence the input reconstruction, similarity preservation, and mutual information across layers. It is shown that the sparser activation of the hidden layer is preferable for preserving information between the input and the output layers. All three models show optimal similarity preservation at dense, not sparse, hidden layer activation. Furthermore, with a large enough hidden layer, it is possible to get zero reconstruction error for any input just by varying the thresholds of neurons. However, we show that the preference for sparsity is due to the noise in the weight matrix between layers. A fixed number of nonzero connections to every neuron achieves better information preservation and input reconstruction for the dense hidden layer activation. The theoretical results give useful insight into models of neural computation based on sparse binary representation and association memory.

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9. Predictive Coding, Variational Autoencoders, and Biological Connections

Joseph Marino

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We present a review of predictive coding, from theoretical neuroscience, and variational autoencoders, from machine learning, identifying the common origin and mathematical framework underlying both areas. As each area is prominent within its respective field, more firmly connecting these areas could prove useful in the dialogue between neuroscience and machine learning. After reviewing each area, we discuss two possible correspondences implied by this perspective: cortical pyramidal dendrites as analogous to (nonlinear) deep networks and lateral inhibition as analogous to normalizing flows. These connections may provide new directions for further investigations in each field.

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10. Tracking Fast and Slow Changes in Synaptic Weights from Simultaneously Observed Pre- and Postsynaptic Spiking

Ganchao Wei, Ian H. Stevenson

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Synapses change on multiple timescales, ranging from milliseconds to minutes, due to a combination of both short- and long-term plasticity. Here we develop an extension of the common generalized linear model to infer both short- and long-term changes in the coupling between a pre- and postsynaptic neuron based on observed spiking activity. We model short-term synaptic plasticity using additive effects that depend on the presynaptic spike timing, and we model long-term changes in both synaptic weight and baseline firing rate using point process adaptive smoothing. Using simulations, we first show that this model can accurately recover time-varying synaptic weights (1) for both depressing and facilitating synapses, (2) with a variety of long-term changes (including realistic changes, such as due to STDP), (3) with a range of pre and postsynaptic firing rates, and (4) for both excitatory and inhibitory synapses. We then apply our model to two experimentally recorded putative synaptic connections. We find that simultaneously tracking fast changes in synaptic weights, slow changes in synaptic weights, and unexplained variations in baseline firing is essential. Omitting any one of these factors can lead to spurious inferences for the others. Altogether, this model provides a flexible framework for tracking short- and long-term variation in spike transmission.

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11. Task-Agnostic Continual Learning Using Online Variational Bayes with Fixed-Point Updates

Chen Zeno, Itay Golan, Elad Hoffer, Daniel Soudry

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Catastrophic forgetting is the notorious vulnerability of neural networks to the changes in the data distribution during learning. This phenomenon has long been considered a major obstacle for using learning agents in realistic continual learning settings. A large body of continual learning research assumes that task boundaries are known during training. However, only a few works consider scenarios in which task boundaries are unknown or not well defined: task-agnostic scenarios. The optimal Bayesian solution for this requires an intractable online Bayes update to the weights posterior. We aim to approximate the online Bayes update as accurately as possible. To do so, we derive novel fixed-point equations for the online variational Bayes optimization problem for multivariate gaussian parametric distributions. By iterating the

posterior through these fixed-point equations, we obtain an algorithm (FOO-VB) for continual learning that can handle nonstationary data distribution using a fixed architecture and without using external memory (i.e., without access to previous data). We demonstrate that our method (FOO-VB) outperforms existing methods in task-agnostic scenarios. FOO-VB Pytorch implementation is available at https://github.com/chenzeno/FOO-VB.

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12. Completion of the Infeasible Actions of Others: Goal Inference by Dynamical Invariant

Takuma Torii, Shohei Hidaka

https://doi.org/10.1162/neco_a_01437 | pp. 1-31

To help another person, we need to infer his or her goal and intention and then perform the action that he or she was unable to perform to meet the intended goal. In this study, we investigate a computational mechanism for inferring someone's intention and goal from that person's incomplete action to enable the action to be completed on his or her behalf. As a minimal and idealized motor control task of this type, we analyzed single-link pendulum control tasks by manipulating the underlying goals. By analyzing behaviors generated by multiple types of these tasks, we found that a type of fractal dimension of movements is characteristic of the difference in the underlying motor controllers, which reflect the difference in the underlying goals. To test whether an incomplete action can be completed using this property of the action trajectory, we demonstrated that the simulated pendulum controller can perform an action in the direction of the underlying goal by using the fractal dimension as a criterion for similarity in movements.

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13. Confidence-Controlled Hebbian Learning Efficiently Extracts Category Membership From Stimuli Encoded in View of a Categorization Task

Kevin Berlemont, Jean■Pierre Nadal

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Abstract In experiments on perceptual decision making, individuals learn a categorization task through trial-and-error protocols. We explore the capacity of a decision-making attractor network to learn a categorization task through reward-based, Hebbian-type modifications of the weights incoming from the stimulus encoding layer. For the latter, we assume a standard layer of a large number of stimulus-specific neurons. Within the general framework of Hebbian learning, we have hypothesized that the learning rate is modulated by the reward at each trial. Surprisingly, we find that when the coding layer has been optimized in view of the categorization task, such reward-modulated Hebbian learning (RMHL) fails to extract efficiently the category membership. In previous work, we showed that the attractor neural networks' nonlinear dynamics accounts for behavioral confidence in sequences of decision trials. Taking advantage of these findings, we propose that learning is controlled by confidence, as computed from the neural activity of the decision-making attractor network. Here we show that this confidence-controlled, reward-based Hebbian learning efficiently extracts categorical information from the optimized coding layer. The proposed learning rule is local and, in contrast to RMHL, does not require storing the average rewards obtained on previous trials. In addition, we find that the confidence-controlled learning rule achieves near-optimal performance. In accordance with this result, we show that the learning rule approximates a gradient descent method on a maximizing reward cost function.

14. Traveling Waves in Quasi-One-Dimensional Neuronal Minicolumns

Vincent Baker, Luis Cruz

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Traveling waves of neuronal activity in the cortex have been observed in vivo. These traveling waves have been correlated to various features of observed cortical dynamics, including spike timing variability and correlated fluctuations in neuron membrane potential. Although traveling waves are typically studied as either strictly one-dimensional or two-dimensional excitations, here we investigate the conditions for the existence of quasi-one-dimensional traveling waves that could be sustainable in parts of the brain containing cortical minicolumns. For that, we explore a quasi-one-dimensional network of heterogeneous neurons with a biologically influenced computational model of neuron dynamics and connectivity. We find that background stimulus reliably evokes traveling waves in networks with local connectivity between neurons. We also observe traveling waves in fully connected networks when a model for action potential propagation speed is incorporated. The biological properties of the neurons influence the generation and propagation of the traveling waves. Our quasi-one-dimensional model is not only useful for studying the basic properties of traveling waves in neuronal networks; it also provides a simplified representation of possible wave propagation in columnar or minicolumnar networks found in the cortex.

15. Bridging the Functional and Wiring Properties of V1 Neurons Through Sparse Coding

Xiaolin Hu, Zhigang Zeng

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The functional properties of neurons in the primary visual cortex (V1) are thought to be closely related to the structural properties of this network, but the specific relationships remain unclear. Previous theoretical studies have suggested that sparse coding, an energy-efficient coding method, might underlie the orientation selectivity of V1 neurons. We thus aimed to delineate how the neurons are wired to produce this feature. We constructed a model and endowed it with a simple Hebbian learning rule to encode images of natural scenes. The excitatory neurons fired sparsely in response to images and developed strong orientation selectivity. After learning, the connectivity between excitatory neuron pairs, inhibitory neuron pairs, and excitatory-inhibitory neuron pairs depended on firing pattern and receptive field similarity between the neurons. The receptive fields (RFs) of excitatory neurons and inhibitory neurons were well predicted by the RFs of presynaptic excitatory neurons and inhibitory neurons, respectively. The excitatory neurons formed a small-world network, in which certain local connection patterns were significantly overrepresented. Bidirectionally manipulating the firing rates of inhibitory neurons caused linear transformations of the firing rates of excitatory neurons, and vice versa. These wiring properties and modulatory effects were congruent with a wide variety of data measured in V1, suggesting that the sparse coding principle might underlie both the functional and wiring properties of V1 neurons.

16. Modeling the Ventral and Dorsal Cortical Visual Pathways Using Artificial Neural Networks

Zhixian Han, Anne B. Sereno
https://doi.org/10.1162/neco_a_01456 | pp. 138-171

Although in conventional models of cortical processing, object recognition and spatial properties are processed separately in ventral and dorsal cortical visual pathways respectively, some recent studies have shown that representations associated with both objects' identity (of shape) and space are present in both visual pathways. However, it is still unclear whether the presence of identity and spatial properties in both pathways have functional roles. In our study, we have tried to answer this question through computational modeling. Our simulation results show that both a model ventral and dorsal pathway, separately trained to do object and spatial recognition, respectively, each actively retained information about both identity and space. In addition, we show that these networks retained different amounts and kinds of identity and spatial information. As a result, our modeling suggests that two separate cortical visual pathways for identity and space (1) actively retain information about both identity and space (2) retain information about identity and space differently and (3) that this differently retained information about identity and space in the two pathways may be necessary to accurately and optimally recognize and localize objects. Further, modeling results suggests these findings are robust and do not strongly depend on the specific structures of the neural networks.

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17. Toward a Brain-Inspired Developmental Neural Network Based on Dendritic Spine Dynamics

Feifei Zhao, Yi Zeng, Jun Bai
https://doi.org/10.1162/neco_a_01448 | pp. 172-189

Neural networks with a large number of parameters are prone to overfitting problems when trained on a relatively small training set. Introducing weight penalties of regularization is a promising technique for solving this problem. Taking inspiration from the dynamic plasticity of dendritic spines, which plays an important role in the maintenance of memory, this letter proposes a brain-inspired developmental neural network based on dendritic spine dynamics (BDNN-dsd). The dynamic structure changes of dendritic spines include appearing, enlarging, shrinking, and disappearing. Such spine plasticity depends on synaptic activity and can be modulated by experiences-in particular, long-lasting synaptic enhancement/suppression (LTP/LTD), coupled with synapse formation (or enlargement)/elimination (or shrinkage), respectively. Subsequently, spine density characterizes an approximate estimate of the total number of synapses between neurons. Motivated by this, we constrain the weight to a tunable bound that can be adaptively modulated based on synaptic activity. Dynamic weight bound could limit the relatively redundant synapses and facilitate the contributing synapses. Extensive experiments demonstrate the effectiveness of our method on classification tasks of different complexity with the MNIST, Fashion MNIST, and CIFAR-10 data sets. Furthermore, compared to dropout and L2 regularization algorithms, our method can improve the network convergence rate and classification performance even for a compact network.

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18. Spatial Attention Enhances Crowded Stimulus Encoding Across Modeled Receptive Fields by Increasing Redundancy of Feature Representations

Justin Theiss, Joel D. Bowen, Michael A. Silver
https://doi.org/10.1162/neco_a_01447 | pp. 190-218

Any visual system, biological or artificial, must make a trade-off between the number of units used to represent the visual environment and the spatial resolution of the sampling array. Humans and some other animals are able to allocate attention to spatial locations to reconfigure the sampling array of receptive fields (RFs), thereby enhancing the spatial resolution of representations without changing the overall number of sampling units. Here, we examine how representations of visual features in a fully convolutional neural network interact and interfere with each other in an eccentricity-dependent RF pooling array and how these interactions are influenced by dynamic changes in spatial resolution across the array. We study these feature interactions within the framework of visual crowding, a well-characterized perceptual phenomenon in which target objects in the visual periphery that are easily identified in isolation are much more difficult to identify when flanked by similar nearby objects. By separately simulating effects of spatial attention on RF size and on the density of the pooling array, we demonstrate that the increase in RF density due to attention is more beneficial than changes in RF size for enhancing target classification for crowded stimuli. Furthermore, by varying target/flanker spacing, as well as the spatial extent of attention, we find that feature redundancy across RFs has more influence on target classification than the fidelity of the feature representations themselves. Based on these findings, we propose a candidate mechanism by which spatial attention relieves visual crowding through enhanced feature redundancy that is mostly due to increased RF density.

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19. A Double-Layer Multi-Resolution Classification Model for Decoding Spatiotemporal Patterns of Spikes With Small Sample Size

Xiwei Shé, Theodore W. Berger, Dong Song

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We build a double-layer, multiple temporal-resolution classification model for decoding single-trial spatiotemporal patterns of spikes. The model takes spiking activities as input signals and binary behavioral or cognitive variables as output signals and represents the input-output mapping with a double-layer ensemble classifier. In the first layer, to solve the underdetermined problem caused by the small sample size and the very high dimensionality of input signals, B-spline functional expansion and L1-regularized logistic classifiers are used to reduce dimensionality and yield sparse model estimations. A wide range of temporal resolutions of neural features is included by using a large number of classifiers with different numbers of B-spline knots. Each classifier serves as a base learner to classify spatiotemporal patterns into the probability of the output label with a single temporal resolution. A bootstrap aggregating strategy is used to reduce the estimation variances of these classifiers. In the second layer, another L1-regularized logistic classifier takes outputs of first-layer classifiers as inputs to generate the final output predictions. This classifier serves as a meta-learner that fuses multiple temporal resolutions to classify spatiotemporal patterns of spikes into binary output labels. We test this decoding model with both synthetic and experimental data recorded from rats and human subjects performing memory-dependent behavioral tasks. Results show that this method can effectively avoid overfitting and yield accurate prediction of output labels with small sample size. The double-layer, multi-resolution classifier consistently outperforms the best single-layer, single-resolution classifier by extracting and utilizing multi-resolution spatiotemporal features of spike patterns in the classification.

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20. Learning in Volatile Environments With the Bayes Factor Surprise

Vasiliki Liakoni, Alireza Modirshanechi, Wulfram Gerstner, Johanni Brea

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Surprise-based learning allows agents to rapidly adapt to nonstationary stochastic environments characterized by sudden changes. We show that exact Bayesian inference in a hierarchical model gives rise to a surprise-modulated trade-off between forgetting old observations and integrating them with the new ones. The modulation depends on a probability ratio, which we call the Bayes Factor Surprise, that tests the prior belief against the current belief. We demonstrate that in several existing approximate algorithms, the Bayes Factor Surprise modulates the rate of adaptation to new observations. We derive three novel surprise-based algorithms, one in the family of particle filters, one in the family of variational learning, and one in the family of message passing, that have constant scaling in observation sequence length and particularly simple update dynamics for any distribution in the exponential family. Empirical results show that these surprise-based algorithms estimate parameters better than alternative approximate approaches and reach levels of performance comparable to computationally more expensive algorithms. The Bayes Factor Surprise is related to but different from the Shannon Surprise. In two hypothetical experiments, we make testable predictions for physiological indicators that dissociate the Bayes Factor Surprise from the Shannon Surprise. The theoretical insight of casting various approaches as surprise-based learning, as well as the proposed online algorithms, may be applied to the analysis of animal and human behavior and to reinforcement learning in nonstationary environments.

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21. Bridging the Gap Between Neurons and Cognition Through Assemblies of Neurons

Christos H. Papadimitriou, Angela D. Friederici

https://doi.org/10.1162/neco_a_01463 | pp. 291-306

During recent decades, our understanding of the brain has advanced dramatically at both the cellular and molecular levels and at the cognitive neurofunctional level; however, a huge gap remains between the microlevel of physiology and the macrolevel of cognition. We propose that computational models based on assemblies of neurons can serve as a blueprint for bridging these two scales. We discuss recently developed computational models of assemblies that have been demonstrated to mediate higher cognitive functions such as the processing of simple sentences, to be realistically realizable by neural activity, and to possess general computational power.

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22. A Normative Account of Confirmation Bias During Reinforcement Learning

Germain Lefebvre, Christopher Summerfield, Rafa■ Bogacz

https://doi.org/10.1162/neco_a_01455 | pp. 307-337

Reinforcement learning involves updating estimates of the value of states and actions on the basis of experience. Previous work has shown that in humans, reinforcement learning exhibits a confirmatory bias: when the value of a chosen option is being updated, estimates are revised more radically following positive than negative reward prediction errors, but the converse is observed when updating the unchosen option

value estimate. Here, we simulate performance on a multi-arm bandit task to examine the consequences of a confirmatory bias for reward harvesting. We report a paradoxical finding: that confirmatory biases allow the agent to maximize reward relative to an unbiased updating rule. This principle holds over a wide range of experimental settings and is most influential when decisions are corrupted by noise. We show that this occurs because on average, confirmatory biases lead to overestimating the value of more valuable bandits and underestimating the value of less valuable bandits, rendering decisions overall more robust in the face of noise. Our results show how apparently suboptimal learning rules can in fact be reward maximizing if decisions are made with finite computational precision.

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23. Model Based or Model Free? Comparing Adaptive Methods for Estimating Thresholds in Neuroscience

Julien Audiffren, Jean■Pierre Bresciani

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The quantification of human perception through the study of psychometric functions Ψ is one of the pillars of experimental psychophysics. In particular, the evaluation of the threshold is at the heart of many neuroscience and cognitive psychology studies, and a wide range of adaptive procedures has been developed to improve its estimation. However, these procedures are often implicitly based on different mathematical assumptions on the psychometric function, and unfortunately, these assumptions cannot always be validated prior to data collection. This raises questions about the accuracy of the estimator produced using the different procedures. In the study we examine in this letter, we compare five adaptive procedures commonly used in psychophysics to estimate the threshold: Dichotomous Optimistic Search (DOS), Staircase, PsiMethod, Gaussian Processes, and QuestPlus. These procedures range from model-based methods, such as the PsiMethod, which relies on strong assumptions regarding the shape of Ψ, to model-free methods, such as DOS, for which assumptions are minimal. The comparisons are performed using simulations of multiple experiments, with psychometric functions of various complexity. The results show that while model-based methods perform well when Ψ is an ideal psychometric function, model-free methods rapidly outshine them when Ψ deviates from this model, as, for instance, when Ψ is a beta cumulative distribution function. Our results highlight the importance of carefully choosing the most appropriate method depending on the context.

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24. Implicit Contact Dynamics Modeling With Explicit Inertia Matrix Representation for Real-Time, Model-Based Control in Physical Environment

Takeshi Itoh, Koji Ishihara, Jun Morimoto

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Model-based control has great potential for use in real robots due to its high sampling efficiency. Nevertheless, dealing with physical contacts and generating accurate motions are inevitable for practical robot control tasks, such as precise manipulation. For a real-time, model-based approach, the difficulty of contact-rich tasks that requires precise movement lies in the fact that a model needs to accurately predict forthcoming contact events within a limited length of time rather than detect them afterward with sensors. Therefore, in this study, we investigate whether and how neural network models can learn a task-related model useful enough for model-based control, that is, a model predicting future states, including contact

events. To this end, we propose a structured neural network model predictive control (SNN-MPC) method, whose neural network architecture is designed with explicit inertia matrix representation. To train the proposed network, we develop a two-stage modeling procedure for contact-rich dynamics from a limited number of samples. As a contact-rich task, we take up a trackball manipulation task using a physical 3-DoF finger robot. The results showed that the SNN-MPC outperformed MPC with a conventional fully connected network model on the manipulation task.

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25. Predicting the Ease of Human Category Learning Using Radial Basis Function Networks

Brett D. Roads, Michael C. Mozer

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Abstract Our goal is to understand and optimize human concept learning by predicting the ease of learning of a particular exemplar or category. We propose a method for estimating ease values, quantitative measures of ease of learning, as an alternative to conducting costly empirical training studies. Our method combines a psychological embedding of domain exemplars with a pragmatic categorization model. The two components are integrated using a radial basis function network (RBFN) that predicts ease values. The free parameters of the RBFN are fit using human similarity judgments, circumventing the need to collect human training data to fit more complex models of human categorization. We conduct two category-training experiments to validate predictions of the RBFN. We demonstrate that an instance-based RBFN outperforms both a prototype-based RBFN and an empirical approach using the raw data. Although the human data were collected across diverse experimental conditions, the predicted ease values strongly correlate with human learning performance. Training can be sequenced by (predicted) ease, achieving what is known as fading in the psychology literature and curriculum learning in the machine-learning literature, both of which have been shown to facilitate learning.

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26. A Neurodynamic Model of Saliency Prediction in V1

David Berga, Xavier Otazu

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Lateral connections in the primary visual cortex (V1) have long been hypothesized to be responsible for several visual processing mechanisms such as brightness induction, chromatic induction, visual discomfort, and bottom-up visual attention (also named saliency). Many computational models have been developed to independently predict these and other visual processes, but no computational model has been able to reproduce all of them simultaneously. In this work, we show that a biologically plausible computational model of lateral interactions of V1 is able to simultaneously predict saliency and all the aforementioned visual processes. Our model's architecture (NSWAM) is based on Penacchio's neurodynamic model of lateral connections of V1. It is defined as a network of firing rate neurons, sensitive to visual features such as brightness, color, orientation, and scale. We tested NSWAM saliency predictions using images from several eye tracking data sets. We show that the accuracy of predictions obtained by our architecture, using shuffled metrics, is similar to other state-of-the-art computational methods, particularly with synthetic images (CAT2000-Pattern and SID4VAM) that mainly contain low-level features. Moreover, we outperform other biologically inspired saliency models that are specifically designed to exclusively reproduce saliency. We

show that our biologically plausible model of lateral connections can simultaneously explain different visual processes present in V1 (without applying any type of training or optimization and keeping the same parameterization for all the visual processes). This can be useful for the definition of a unified architecture of the primary visual cortex.

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27. Distributed Phase Oscillatory Excitation Efficiently Produces Attractors Using Spike-Timing-Dependent Plasticity

Eric C. Wong

https://doi.org/10.1162/neco_a_01466 | pp. 415-436

The brain is thought to represent information in the form of activity in distributed groups of neurons known as attractors. We show here that in a randomly connected network of simulated spiking neurons, periodic stimulation of neurons with distributed phase offsets, along with standard spike-timing-dependent plasticity (STDP), efficiently creates distributed attractors. These attractors may have a consistent ordered firing pattern or become irregular, depending on the conditions. We also show that when two such attractors are stimulated in sequence, the same STDP mechanism can create a directed association between them, forming the basis of an associative network. We find that for an STDP time constant of 20 ms, the dependence of the efficiency of attractor creation on the driving frequency has a broad peak centered around 8 Hz. Upon restimulation, the attractors self-oscillate, but with an oscillation frequency that is higher than the driving frequency, ranging from 10 to 100 Hz.

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28. Categorical Perception: A Groundwork for Deep Learning

Laurent Bonnasse■Gahot, Jean■Pierre Nadal

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Classification is one of the major tasks that deep learning is successfully tackling. Categorization is also a fundamental cognitive ability. A well-known perceptual consequence of categorization in humans and other animals, categorical perception, is notably characterized by a within-category compression and a between-category separation: two items, close in input space, are perceived closer if they belong to the same category than if they belong to different categories. Elaborating on experimental and theoretical results in cognitive science, here we study categorical effects in artificial neural networks. We combine a theoretical analysis that makes use of mutual and Fisher information quantities and a series of numerical simulations on networks of increasing complexity. These formal and numerical analyses provide insights into the geometry of the neural representation in deep layers, with expansion of space near category boundaries and contraction far from category boundaries. We investigate categorical representation by using two complementary approaches: one mimics experiments in psychophysics and cognitive neuroscience by means of morphed continua between stimuli of different categories, while the other introduces a categoricality index that, for each layer in the network, quantifies the separability of the categories at the neural population level. We show on both shallow and deep neural networks that category learning automatically induces categorical perception. We further show that the deeper a layer, the stronger the categorical effects. As an outcome of our study, we propose a coherent view of the efficacy of different heuristic practices of the dropout regularization technique. More generally, our view, which finds echoes in the neuroscience literature, insists on the differential impact of noise in any given layer depending on the

geometry of the neural representation that is being learned, that is, on how this geometry reflects the structure of the categories.

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29. Whence the Expected Free Energy?

Beren Millidge, Alexander Tschantz, Christopher L. Buckley

https://doi.org/10.1162/neco_a_01354 | pp. 447-482

The expected free energy (EFE) is a central quantity in the theory of active inference. It is the quantity that all active inference agents are mandated to minimize through action, and its decomposition into extrinsic and intrinsic value terms is key to the balance of exploration and exploitation that active inference agents evince. Despite its importance, the mathematical origins of this quantity and its relation to the variational free energy (VFE) remain unclear. In this letter, we investigate the origins of the EFE in detail and show that it is not simply "the free energy in the future." We present a functional that we argue is the natural extension of the VFE but actively discourages exploratory behavior, thus demonstrating that exploration does not directly follow from free energy minimization into the future. We then develop a novel objective, the free energy of the expected future (FEEF), which possesses both the epistemic component of the EFE and an intuitive mathematical grounding as the divergence between predicted and desired futures.

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30. Nonconvex Sparse Regularization for Deep Neural Networks and Its Optimality

Ilsang Ohn, Yongdai Kim

https://doi.org/10.1162/neco_a_01457 | pp. 476-517

Recent theoretical studies proved that deep neural network (DNN) estimators obtained by minimizing empirical risk with a certain sparsity constraint can attain optimal convergence rates for regression and classification problems. However, the sparsity constraint requires knowing certain properties of the true model, which are not available in practice. Moreover, computation is difficult due to the discrete nature of the sparsity constraint. In this letter, we propose a novel penalized estimation method for sparse DNNs that resolves the problems existing in the sparsity constraint. We establish an oracle inequality for the excess risk of the proposed sparse-penalized DNN estimator and derive convergence rates for several learning tasks. In particular, we prove that the sparse-penalized estimator can adaptively attain minimax convergence rates for various nonparametric regression problems. For computation, we develop an efficient gradient-based optimization algorithm that guarantees the monotonic reduction of the objective function.

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31. Convolution-Based Model-Solving Method for Three-Dimensional, Unsteady, Partial Differential Equations

Wenshu Zha, Wen Zhang, Daolun Li, Yan Xing, Lei He, Jieqing Tan

https://doi.org/10.1162/neco_a_01462 | pp. 518-540

Neural networks are increasingly used widely in the solution of partial differential equations (PDEs). This letter proposes 3D-PDE-Net to solve the three-dimensional PDE. We give a mathematical derivation of a three-dimensional convolution kernel that can approximate any order differential operator within the range of

expressing ability and then conduct 3D-PDE-Net based on this theory. An optimum network is obtained by minimizing the normalized mean square error (NMSE) of training data, and L-BFGS is the optimized algorithm of second-order precision. Numerical experimental results show that 3D-PDE-Net can achieve the solution with good accuracy using few training samples, and it is of highly significant in solving linear and nonlinear unsteady PDEs.

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32. Stability Conditions of Bicomplex-Valued Hopfield Neural Networks

Masaki Kobayashi

https://doi.org/10.1162/neco_a_01350 | pp. 552-562

Hopfield neural networks have been extended using hypercomplex numbers. The algebra of bicomplex numbers, also referred to as commutative quaternions, is a number system of dimension 4. Since the multiplication is commutative, many notions and theories of linear algebra, such as determinant, are available, unlike quaternions. A bicomplex-valued Hopfield neural network (BHNN) has been proposed as a multistate neural associative memory. However, the stability conditions have been insufficient for the projection rule. In this work, the stability conditions are extended and applied to improvement of the projection rule. The computer simulations suggest improved noise tolerance.

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33. From Biophysical to Integrate-and-Fire Modeling

Tomas Van Pottelbergh, Guillaume Drion, Rodolphe Sepulchre

https://doi.org/10.1162/neco_a_01353 | pp. 563-589

This article proposes a methodology to extract a low-dimensional integrate-and-fire model from an arbitrarily detailed single-compartment biophysical model. The method aims at relating the modulation of maximal conductance parameters in the biophysical model to the modulation of parameters in the proposed integrate-and-fire model. The approach is illustrated on two well-documented examples of cellular neuromodulation: the transition between type I and type II excitability and the transition between spiking and bursting.

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34. Implicit Regularization and Momentum Algorithms in Nonlinearly Parameterized Adaptive Control and Prediction

Nicholas M. Boffi, Jean-Jacques Slotine

https://doi.org/10.1162/neco_a_01360 | pp. 590-673

Stable concurrent learning and control of dynamical systems is the subject of adaptive control. Despite being an established field with many practical applications and a rich theory, much of the development in adaptive control for nonlinear systems revolves around a few key algorithms. By exploiting strong connections between classical adaptive nonlinear control techniques and recent progress in optimization and machine learning, we show that there exists considerable untapped potential in algorithm development for both adaptive nonlinear control and adaptive dynamics prediction. We begin by introducing first-order adaptation laws inspired by natural gradient descent and mirror descent. We prove that when there are multiple dynamics consistent with the data, these non-Euclidean adaptation laws implicitly regularize the learned

model. Local geometry imposed during learning thus may be used to select parameter vectors-out of the many that will achieve perfect tracking or prediction-for desired properties such as sparsity. We apply this result to regularized dynamics predictor and observer design, and as concrete examples, we consider Hamiltonian systems, Lagrangian systems, and recurrent neural networks. We subsequently develop a variational formalism based on the Bregman Lagrangian. We show that its Euler Lagrange equations lead to natural gradient and mirror descent-like adaptation laws with momentum, and we recover their first-order analogues in the infinite friction limit. We illustrate our analyses with simulations demonstrating our theoretical results.

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35. Active Inference: Demystified and Compared

Noor Sajid, Philip J. Ball, Thomas Parr, Karl J. Friston

https://doi.org/10.1162/neco_a_01357 | pp. 674-712

Active inference is a first principle account of how autonomous agents operate in dynamic, nonstationary environments. This problem is also considered in reinforcement learning, but limited work exists on comparing the two approaches on the same discrete-state environments. In this letter, we provide (1) an accessible overview of the discrete-state formulation of active inference, highlighting natural behaviors in active inference that are generally engineered in reinforcement learning, and (2) an explicit discrete-state comparison between active inference and reinforcement learning on an OpenAI gym baseline. We begin by providing a condensed overview of the active inference literature, in particular viewing the various natural behaviors of active inference agents through the lens of reinforcement learning. We show that by operating in a pure belief-based setting, active inference agents can carry out epistemic exploration-and account for uncertainty about their environment-in a Bayes-optimal fashion. Furthermore, we show that the reliance on an explicit reward signal in reinforcement learning is removed in active inference, where reward can simply be treated as another observation we have a preference over; even in the total absence of rewards, agent behaviors are learned through preference learning. We make these properties explicit by showing two scenarios in which active inference agents can infer behaviors in reward-free environments compared to both Q-learning and Bayesian model-based reinforcement learning agents and by placing zero prior preferences over rewards and learning the prior preferences over the observations corresponding to reward. We conclude by noting that this formalism can be applied to more complex settings (e.g., robotic arm movement, Atari games) if appropriate generative models can be formulated. In short, we aim to demystify the behavior of active inference agents by presenting an accessible discrete state-space and time formulation and demonstrate these behaviors in a OpenAI gym environment, alongside reinforcement learning agents.

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36. Sophisticated Inference

Karl Friston, Lancelot Da Costa, Danijar Hafner, Casper Hesp, Thomas Parr

https://doi.org/10.1162/neco_a_01351 | pp. 713-763

Active inference offers a first principle account of sentient behavior, from which special and important cases—for example, reinforcement learning, active learning, Bayes optimal inference, Bayes optimal design—can be derived. Active inference finesses the exploitation-exploration dilemma in relation to prior preferences by placing information gain on the same footing as reward or value. In brief, active inference

replaces value functions with functionals of (Bayesian) beliefs, in the form of an expected (variational) free energy. In this letter, we consider a sophisticated kind of active inference using a recursive form of expected free energy. Sophistication describes the degree to which an agent has beliefs about beliefs. We consider agents with beliefs about the counterfactual consequences of action for states of affairs and beliefs about those latent states. In other words, we move from simply considering beliefs about "what would happen if I did that" to "what I would believe about what would happen if I did that." The recursive form of the free energy functional effectively implements a deep tree search over actions and outcomes in the future. Crucially, this search is over sequences of belief states as opposed to states per se. We illustrate the competence of this scheme using numerical simulations of deep decision problems.

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37. Statistical Analysis of Decoding Performances of Diverse Populations of Neurons

Kyle Wendling, Cheng Ly

https://doi.org/10.1162/neco_a_01355 | pp. 764-801

A central theme in computational neuroscience is determining the neural correlates of efficient and accurate coding of sensory signals. Diversity, or heterogeneity, of intrinsic neural attributes is known to exist in many brain areas and is thought to significantly affect neural coding. Recent theoretical and experimental work has argued that in uncoupled networks, coding is most accurate at intermediate levels of heterogeneity. Here we consider this question with data from in vivo recordings of neurons in the electrosensory system of weakly electric fish subject to the same realization of noisy stimuli; we use a generalized linear model (GLM) to assess the accuracy of (Bayesian) decoding of stimulus given a population spiking response. The long recordings enable us to consider many uncoupled networks and a relatively wide range of heterogeneity, as well as many instances of the stimuli, thus enabling us to address this question with statistical power. The GLM decoding is performed on a single long time series of data to mimic realistic conditions rather than using trial-averaged data for better model fits. For a variety of fixed network sizes, we generally find that the optimal levels of heterogeneity are at intermediate values, and this holds in all core components of GLM. These results are robust to several measures of decoding performance, including the absolute value of the error, error weighted by the uncertainty of the estimated stimulus, and the correlation between the actual and estimated stimulus. Although a quadratic fit to decoding performance as a function of heterogeneity is statistically significant, the result is highly variable with low [Formula: see text] values. Taken together, intermediate levels of neural heterogeneity are indeed a prominent attribute for efficient coding even within a single time series, but the performance is highly variable.

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38. Unsupervised Discovery, Control, and Disentanglement of Semantic Attributes With Applications to Anomaly Detection

William Paul, I-Jeng Wang, Fady Alajaji, Philippe Burlina

https://doi.org/10.1162/neco_a_01359 | pp. 802-826

Our work focuses on unsupervised and generative methods that address the following goals: (1) learning unsupervised generative representations that discover latent factors controlling image semantic attributes, (2) studying how this ability to control attributes formally relates to the issue of latent factor disentanglement, clarifying related but dissimilar concepts that had been confounded in the past, and (3) developing anomaly

detection methods that leverage representations learned in the first goal. For goal 1, we propose a network architecture that exploits the combination of multiscale generative models with mutual information (MI) maximization. For goal 2, we derive an analytical result, lemma 1, that brings clarity to two related but distinct concepts: the ability of generative networks to control semantic attributes of images they generate, resulting from MI maximization, and the ability to disentangle latent space representations, obtained via total correlation minimization. More specifically, we demonstrate that maximizing semantic attribute control encourages disentanglement of latent factors. Using lemma 1 and adopting MI in our loss function, we then show empirically that for image generation tasks, the proposed approach exhibits superior performance as measured in the quality and disentanglement of the generated images when compared to other state-of-the-art methods, with quality assessed via the Fréchet inception distance (FID) and disentanglement via mutual information gap. For goal 3, we design several systems for anomaly detection exploiting representations learned in goal 1 and demonstrate their performance benefits when compared to state-of-the-art generative and discriminative algorithms. Our contributions in representation learning have potential applications in addressing other important problems in computer vision, such as bias and privacy in AI.

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39. Mapping Low-Dimensional Dynamics to High-Dimensional Neural Activity: A Derivation of the Ring Model From the Neural Engineering Framework

Omri Barak, Sandro Romani

https://doi.org/10.1162/neco_a_01361 | pp. 827-852

Empirical estimates of the dimensionality of neural population activity are often much lower than the population size. Similar phenomena are also observed in trained and designed neural network models. These experimental and computational results suggest that mapping low-dimensional dynamics to high-dimensional neural space is a common feature of cortical computation. Despite the ubiquity of this observation, the constraints arising from such mapping are poorly understood. Here we consider a specific example of mapping low-dimensional dynamics to high-dimensional neural activity-the neural engineering framework. We analytically solve the framework for the classic ring model-a neural network encoding a static or dynamic angular variable. Our results provide a complete characterization of the success and failure modes for this model. Based on similarities between this and other frameworks, we speculate that these results could apply to more general scenarios.

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40. The Effect of Class Imbalance on Precision-Recall Curves

Christopher K. I. Williams

https://doi.org/10.1162/neco_a_01362 | pp. 853-857

In this note, I study how the precision of a binary classifier depends on the ratio r of positive to negative cases in the test set, as well as the classifier's true and false-positive rates. This relationship allows prediction of how the precision-recall curve will change with r, which seems not to be well known. It also allows prediction of how Fβ and the precision gain and recall gain measures of Flach and Kull (2015) vary with r.

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41. How Convolutional Neural Network Architecture Biases Learned Opponency and Color Tuning

Ethan Harris, Daniela Mihai, Jonathon Hare

https://doi.org/10.1162/neco_a_01356 | pp. 858-898

Recent work suggests that changing convolutional neural network (CNN) architecture by introducing a bottleneck in the second layer can yield changes in learned function. To understand this relationship fully requires a way of quantitatively comparing trained networks. The fields of electrophysiology and psychophysics have developed a wealth of methods for characterizing visual systems that permit such comparisons. Inspired by these methods, we propose an approach to obtaining spatial and color tuning curves for convolutional neurons that can be used to classify cells in terms of their spatial and color opponency. We perform these classifications for a range of CNNs with different depths and bottleneck widths. Our key finding is that networks with a bottleneck show a strong functional organization: almost all cells in the bottleneck layer become both spatially and color opponent, and cells in the layer following the bottleneck become nonopponent. The color tuning data can further be used to form a rich understanding of how color a network encodes color. As a concrete demonstration, we show that shallower networks without a bottleneck learn a complex nonlinear color system, whereas deeper networks with tight bottlenecks learn a simple channel opponent code in the bottleneck layer. We develop a method of obtaining a hue sensitivity curve for a trained CNN that enables high-level insights that complement the low-level findings from the color tuning data. We go on to train a series of networks under different conditions to ascertain the robustness of the discussed results. Ultimately our methods and findings coalesce with prior art, strengthening our ability to interpret trained CNNs and furthering our understanding of the connection between architecture and learned representation. Trained models and code for all experiments are available at https://github.com/ecs-vlc/opponency.

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42. The Remarkable Robustness of Surrogate Gradient Learning for Instilling Complex Function in Spiking Neural Networks

Friedemann Zenke, Tim P. Vogels

https://doi.org/10.1162/neco_a_01367 | pp. 899-925

Brains process information in spiking neural networks. Their intricate connections shape the diverse functions these networks perform. Yet how network connectivity relates to function is poorly understood, and the functional capabilities of models of spiking networks are still rudimentary. The lack of both theoretical insight and practical algorithms to find the necessary connectivity poses a major impediment to both studying information processing in the brain and building efficient neuromorphic hardware systems. The training algorithms that solve this problem for artificial neural networks typically rely on gradient descent. But doing so in spiking networks has remained challenging due to the nondifferentiable nonlinearity of spikes. To avoid this issue, one can employ surrogate gradients to discover the required connectivity. However, the choice of a surrogate is not unique, raising the question of how its implementation influences the effectiveness of the method. Here, we use numerical simulations to systematically study how essential design parameters of surrogate gradients affect learning performance on a range of classification problems. We show that surrogate gradient learning is robust to different shapes of underlying surrogate derivatives, but the choice of the derivative's scale can substantially affect learning performance. When we combine surrogate gradients with suitable activity regularization techniques, spiking networks perform robust

information processing at the sparse activity limit. Our study provides a systematic account of the remarkable robustness of surrogate gradient learning and serves as a practical guide to model functional spiking neural networks.

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43. Flexible Frequency Switching in Adult Mouse Visual Cortex Is Mediated by Competition Between Parvalbumin and Somatostatin Expressing Interneurons

Justin W. M. Domhof, Paul Tiesinga

https://doi.org/10.1162/neco_a_01369 | pp. 926-966

Neuronal networks in rodent primary visual cortex (V1) can generate oscillations in different frequency bands depending on the network state and the level of visual stimulation. High-frequency gamma rhythms, for example, dominate the network's spontaneous activity in adult mice but are attenuated upon visual stimulation, during which the network switches to the beta band instead. The spontaneous local field potential (LFP) of juvenile mouse V1, however, mainly contains beta rhythms and presenting a stimulus does not elicit drastic changes in network oscillations. We study, in a spiking neuron network model, the mechanism in adult mice allowing for flexible switches between multiple frequency bands and contrast this to the network structure in juvenile mice that lack this flexibility. The model comprises excitatory pyramidal cells (PCs) and two types of interneurons: the parvalbumin-expressing (PV) and the somatostatinexpressing (SOM) interneuron. In accordance with experimental findings, the pyramidal-PV and pyramidal-SOM cell subnetworks are associated with gamma and beta oscillations, respectively. In our model, they are both generated via a pyramidal-interneuron gamma (PING) mechanism, wherein the PCs drive the oscillations. Furthermore, we demonstrate that large but not small visual stimulation activates SOM cells, which shift the frequency of resting-state gamma oscillations produced by the pyramidal-PV cell subnetwork so that beta rhythms emerge. Finally, we show that this behavior is obtained for only a subset of PV and SOM interneuron projection strengths, indicating that their influence on the PCs should be balanced so that they can compete for oscillatory control of the PCs. In sum, we propose a mechanism by which visual beta rhythms can emerge from spontaneous gamma oscillations in a network model of the mouse V1; for this mechanism to reproduce V1 dynamics in adult mice, balance between the effective strengths of PV and SOM cells is required.

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44. Real-Time Decoding of Attentional States Using Closed-Loop EEG Neurofeedback

Greta Tuckute, Sofie Therese Hansen, Troels W. Kjær, Lars Kai Hansen

https://doi.org/10.1162/neco_a_01363 | pp. 967-1004

Sustained attention is a cognitive ability to maintain task focus over extended periods of time (Mackworth, 1948; Chun, Golomb, & Turk-Browne, 2011). In this study, scalp electroencephalography (EEG) signals were processed in real time using a 32 dry-electrode system during a sustained visual attention task. An attention training paradigm was implemented, as designed in DeBettencourt, Cohen, Lee, Norman, and Turk-Browne (2015) in which the composition of a sequence of blended images is updated based on the participant's decoded attentional level to a primed image category. It was hypothesized that a single neurofeedback training session would improve sustained attention abilities. Twenty-two participants were

trained on a single neurofeedback session with behavioral pretraining and posttraining sessions within three consecutive days. Half of the participants functioned as controls in a double-blinded design and received sham neurofeedback. During the neurofeedback session, attentional states to primed categories were decoded in real time and used to provide a continuous feedback signal customized to each participant in a closed-loop approach. We report a mean classifier decoding error rate of 34.3% (chance = 50%). Within the neurofeedback group, there was a greater level of task-relevant attentional information decoded in the participant's brain before making a correct behavioral response than before an incorrect response. This effect was not visible in the control group (interaction p=7.23e-4), which strongly indicates that we were able to achieve a meaningful measure of subjective attentional state in real time and control participants' behavior during the neurofeedback session. We do not provide conclusive evidence whether the single neurofeedback session per se provided lasting effects in sustained attention abilities. We developed a portable EEG neurofeedback system capable of decoding attentional states and predicting behavioral choices in the attention task at hand. The neurofeedback code framework is Python based and open source, and it allows users to actively engage in the development of neurofeedback tools for scientific and translational use.

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45. Deep Network With Approximation Error Being Reciprocal of Width to Power of Square Root of Depth

Zuowei Shen, Haizhao Yang, Shijun Zhang

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A new network with super-approximation power is introduced. This network is built with Floor (■x■) or ReLU (max{0,x}) activation function in each neuron; hence, we call such networks Floor-ReLU networks. For any hyperparameters N∈ N+ and L∈ N+, we show that Floor-ReLU networks with width max{d,5N+13} and depth 64dL+3 can uniformly approximate a Hölder function f on [0,1]d with an approximation error 3λdα/2N-αL, where α∈(0,1] and λ are the Hölder order and constant, respectively. More generally for an arbitrary continuous function f on [0,1]d with a modulus of continuity ωf(·), the constructive approximation rate is ωf(dN-L)+2ωf(d)N-L. As a consequence, this new class of networks overcomes the curse of dimensionality in approximation power when the variation of ωf(r) as r→0 is moderate (e.g., ωf(r)■rα for Hölder continuous functions), since the major term to be considered in our approximation rate is essentially d times a function of N and L independent of d within the modulus of continuity.

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46. A Generalization of Spatial Monte Carlo Integration

Muneki Yasuda, Kei Uchizawa

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Spatial Monte Carlo integration (SMCI) is an extension of standard Monte Carlo integration and can approximate expectations on Markov random fields with high accuracy. SMCI was applied to pairwise Boltzmann machine (PBM) learning, achieving superior results over those of some existing methods. The approximation level of SMCI can be altered, and it was proved that a higher-order approximation of SMCI is statistically more accurate than a lower-order approximation. However, SMCI as proposed in previous studies suffers from a limitation that prevents the application of a higher-order method to dense systems. This study makes two contributions. First, a generalization of SMCI (called generalized SMCI (GSMCI)) is

proposed, which allows a relaxation of the above-mentioned limitation; moreover, a statistical accuracy bound of GSMCI is proved. Second, a new PBM learning method based on SMCI is proposed, which is obtained by combining SMCI and persistent contrastive divergence. The proposed learning method significantly improves learning accuracy.

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47. Low-Dimensional Manifolds Support Multiplexed Integrations in Recurrent Neural Networks

Arnaud Fanthomme, Rémi Monasson

https://doi.org/10.1162/neco_a_01366 | pp. 1063-1112

We study the learning dynamics and the representations emerging in recurrent neural networks (RNNs) trained to integrate one or multiple temporal signals. Combining analytical and numerical investigations, we characterize the conditions under which an RNN with n neurons learns to integrate D(■n) scalar signals of arbitrary duration. We show, for linear, ReLU, and sigmoidal neurons, that the internal state lives close to a D-dimensional manifold, whose shape is related to the activation function. Each neuron therefore carries, to various degrees, information about the value of all integrals. We discuss the deep analogy between our results and the concept of mixed selectivity forged by computational neuroscientists to interpret cortical recordings.

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48. Joint Structure and Parameter Optimization of Multiobjective Sparse Neural Network

Junhao Huang, Weize Sun, Lei Huang

https://doi.org/10.1162/neco_a_01368 | pp. 1113-1143

This work addresses the problem of network pruning and proposes a novel joint training method based on a multiobjective optimization model. Most of the state-of-the-art pruning methods rely on user experience for selecting the sparsity ratio of the weight matrices or tensors, and thus suffer from severe performance reduction with inappropriate user-defined parameters. Moreover, networks might be inferior due to the inefficient connecting architecture search, especially when it is highly sparse. It is revealed in this work that the network model might maintain sparse characteristic in the early stage of the backpropagation (BP) training process, and evolutionary computation-based algorithms can accurately discover the connecting architecture with satisfying network performance. In particular, we establish a multiobjective sparse model for network pruning and propose an efficient approach that combines BP training and two modified multiobjective evolutionary algorithms (MOEAs). The BP algorithm converges quickly, and the two MOEAs can search for the optimal sparse structure and refine the weights, respectively. Experiments are also included to prove the benefits of the proposed algorithm. We show that the proposed method can obtain a desired Pareto front (PF), leading to a better pruning result comparing to the state-of-the-art methods, especially when the network structure is highly sparse.

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49. The Refractory Period Matters: Unifying Mechanisms of Macroscopic Brain Waves

Corey Weistuch, Lilianne R. Mujica■Parodi, Ken A. Dill
https://doi.org/10.1162/neco_a_01371 | pp. 1145-1163

The relationship between complex brain oscillations and the dynamics of individual neurons is poorly understood. Here we utilize maximum caliber, a dynamical inference principle, to build a minimal yet general model of the collective (mean field) dynamics of large populations of neurons. In agreement with previous experimental observations, we describe a simple, testable mechanism, involving only a single type of neuron, by which many of these complex oscillatory patterns may emerge. Our model predicts that the refractory period of neurons, which has often been neglected, is essential for these behaviors.

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50. Toward a Kernel-Based Uncertainty Decomposition Framework for Data and Models

Rishabh Singh, José C. Pr■■ncipe
https://doi.org/10.1162/neco_a_01372 | pp. 1164-1198

This letter introduces a new framework for quantifying predictive uncertainty for both data and models that relies on projecting the data into a gaussian reproducing kernel Hilbert space (RKHS) and transforming the data probability density function (PDF) in a way that quantifies the flow of its gradient as a topological potential field (quantified at all points in the sample space). This enables the decomposition of the PDF gradient flow by formulating it as a moment decomposition problem using operators from quantum physics, specifically Schrödinger's formulation. We experimentally show that the higher-order moments systematically cluster the different tail regions of the PDF, thereby providing unprecedented discriminative resolution of data regions having high epistemic uncertainty. In essence, this approach decomposes local realizations of the data PDF in terms of uncertainty moments. We apply this framework as a surrogate tool for predictive uncertainty quantification of point-prediction neural network models, overcoming various limitations of conventional Bayesian-based uncertainty quantification methods. Experimental comparisons with some established methods illustrate performance advantages that our framework exhibits.

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51. Training Spiking Neural Networks in the Strong Coupling Regime

Christopher M. Kim, Carson C. Chow
https://doi.org/10.1162/neco_a_01379 | pp. 1199-1233

Recurrent neural networks trained to perform complex tasks can provide insight into the dynamic mechanism that underlies computations performed by cortical circuits. However, due to a large number of unconstrained synaptic connections, the recurrent connectivity that emerges from network training may not be biologically plausible. Therefore, it remains unknown if and how biological neural circuits implement dynamic mechanisms proposed by the models. To narrow this gap, we developed a training scheme that, in addition to achieving learning goals, respects the structural and dynamic properties of a standard cortical circuit model: strongly coupled excitatory-inhibitory spiking neural networks. By preserving the strong mean excitatory and inhibitory coupling of initial networks, we found that most of trained synapses obeyed Dale's law without additional constraints, exhibited large trial-to-trial spiking variability, and operated in inhibition-stabilized regime. We derived analytical estimates on how training and network parameters constrained the changes in mean synaptic strength during training. Our results demonstrate that training

recurrent neural networks subject to strong coupling constraints can result in connectivity structure and dynamic regime relevant to cortical circuits.

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52. Classification From Pairwise Similarities/Dissimilarities and Unlabeled Data via Empirical Risk Minimization

Takuya Shimada, Han Bao, Issei Sato, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01373 | pp. 1234-1268

Pairwise similarities and dissimilarities between data points are often obtained more easily than full labels of data in real-world classification problems. To make use of such pairwise information, an empirical risk minimization approach has been proposed, where an unbiased estimator of the classification risk is computed from only pairwise similarities and unlabeled data. However, this approach has not yet been able to handle pairwise dissimilarities. Semisupervised clustering methods can incorporate both similarities and dissimilarities into their framework; however, they typically require strong geometrical assumptions on the data distribution such as the manifold assumption, which may cause severe performance deterioration. In this letter, we derive an unbiased estimator of the classification risk based on all of similarities and dissimilarities and unlabeled data. We theoretically establish an estimation error bound and experimentally demonstrate the practical usefulness of our empirical risk minimization method.

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53. Parameter Estimation in Multiple Dynamic Synaptic Coupling Model Using Bayesian Point Process State-Space Modeling Framework

Yalda Amidi, Behzad Nazari, S. Sadri, Ali Yousefi

https://doi.org/10.1162/neco_a_01375 | pp. 1269-1299

It is of great interest to characterize the spiking activity of individual neurons in a cell ensemble. Many different mechanisms, such as synaptic coupling and the spiking activity of itself and its neighbors, drive a cell's firing properties. Though this is a widely studied modeling problem, there is still room to develop modeling solutions by simplifications embedded in previous models. The first shortcut is that synaptic coupling mechanisms in previous models do not replicate the complex dynamics of the synaptic response. The second is that the number of synaptic connections in these models is an order of magnitude smaller than in an actual neuron. In this research, we push this barrier by incorporating a more accurate model of the synapse and propose a system identification solution that can scale to a network incorporating hundreds of synaptic connections. Although a neuron has hundreds of synaptic connections, only a subset of these connections significantly contributes to its spiking activity. As a result, we assume the synaptic connections are sparse, and to characterize these dynamics, we propose a Bayesian point-process state-space model that lets us incorporate the sparsity of synaptic connections within the regularization technique into our framework. We develop an extended expectation-maximization. algorithm to estimate the free parameters of the proposed model and demonstrate the application of this methodology to the problem of estimating the parameters of many dynamic synaptic connections. We then go through a simulation example consisting of the dynamic synapses across a range of parameter values and show that the model parameters can be estimated using our method. We also show the application of the proposed algorithm in the intracellular data that contains 96 presynaptic connections and assess the estimation accuracy of our method using a combination of goodness-of-fit measures.

54. Contrastive Similarity Matching for Supervised Learning

Shanshan Qin, Nayantara Mudur, Cengiz Pehlevan

https://doi.org/10.1162/neco_a_01374 | pp. 1300-1328

We propose a novel biologically plausible solution to the credit assignment problem motivated by observations in the ventral visual pathway and trained deep neural networks. In both, representations of objects in the same category become progressively more similar, while objects belonging to different categories become less similar. We use this observation to motivate a layer-specific learning goal in a deep network: each layer aims to learn a representational similarity matrix that interpolates between previous and later layers. We formulate this idea using a contrastive similarity matching objective function and derive from it deep neural networks with feedforward, lateral, and feedback connections and neurons that exhibit biologically plausible Hebbian and anti-Hebbian plasticity. Contrastive similarity matching can be interpreted as an energy-based learning algorithm, but with significant differences from others in how a contrastive function is constructed.

55. Detecting Scene-Plausible Perceptible Backdoors in Trained DNNs Without Access to the Training Set

Zhen Xiang, David J. Miller, Hang Wang, George Kesidis

https://doi.org/10.1162/neco_a_01376 | pp. 1329-1371

Backdoor data poisoning attacks add mislabeled examples to the training set, with an embedded backdoor pattern, so that the classifier learns to classify to a target class whenever the backdoor pattern is present in a test sample. Here, we address posttraining detection of scene-plausible perceptible backdoors, a type of backdoor attack that can be relatively easily fashioned, particularly against DNN image classifiers. A post-training defender does not have access to the potentially poisoned training set, only to the trained classifier, as well as some unpoisoned examples that need not be training samples. Without the poisoned training set, the only information about a backdoor pattern is encoded in the DNN's trained weights. This detection scenario is of great import considering legacy and proprietary systems, cell phone apps, as well as training outsourcing, where the user of the classifier will not have access to the entire training set. We identify two important properties of scene-plausible perceptible backdoor patterns, spatial invariance and robustness, based on which we propose a novel detector using the maximum achievable misclassification fraction (MAMF) statistic. We detect whether the trained DNN has been backdoor-attacked and infer the source and target classes. Our detector outperforms existing detectors and, coupled with an imperceptible backdoor detector, helps achieve posttraining detection of most evasive backdoors of interest.

56. Hierarchical Dynamical Model for Multiple Cortical Neural Decoding

Xi Liu, Xiang Shen, Shuhang Chen, Xiang Zhang, Yifan Huang, Yueming Wang, Yiwen Wang

https://doi.org/10.1162/neco_a_01380 | pp. 1372-1401

Motor brain machine interfaces (BMIs) interpret neural activities from motor-related cortical areas in the brain into movement commands to control a prosthesis. As the subject adapts to control the neural prosthesis, the medial prefrontal cortex (mPFC), upstream of the primary motor cortex (M1), is heavily

involved in reward-guided motor learning. Thus, considering mPFC and M1 functionality within a hierarchical structure could potentially improve the effectiveness of BMI decoding while subjects are learning. The commonly used Kalman decoding method with only one simple state model may not be able to represent the multiple brain states that evolve over time as well as along the neural pathway. In addition, the performance of Kalman decoders degenerates in heavy-tailed nongaussian noise, which is usually generated due to the nonlinear neural system or influences of movement-related noise in online neural recording. In this letter, we propose a hierarchical model to represent the brain states from multiple cortical areas that evolve along the neural pathway. We then introduce correntropy theory into the hierarchical structure to address the heavy-tailed noise existing in neural recordings. We test the proposed algorithm on in vivo recordings collected from the mPFC and M1 of two rats when the subjects were learning to perform a lever-pressing task. Compared with the classic Kalman filter, our results demonstrate better movement decoding performance due to the hierarchical structure that integrates the past failed trial information over multisite recording and the combination with correntropy criterion to deal with noisy heavy-tailed neural recordings.

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57. Predictive Processing in Cognitive Robotics: A Review

Alejandra Ciria, Guido Schillaci, Giovanni Pezzulo, Verena V. Hafner, Bruno Lara

https://doi.org/10.1162/neco_a_01383 | pp. 1402-1432

Predictive processing has become an influential framework in cognitive sciences. This framework turns the traditional view of perception upside down, claiming that the main flow of information processing is realized in a top-down, hierarchical manner. Furthermore, it aims at unifying perception, cognition, and action as a single inferential process. However, in the related literature, the predictive processing framework and its associated schemes, such as predictive coding, active inference, perceptual inference, and free-energy principle, tend to be used interchangeably. In the field of cognitive robotics, there is no clear-cut distinction on which schemes have been implemented and under which assumptions. In this letter, working definitions are set with the main aim of analyzing the state of the art in cognitive robotics research working under the predictive processing framework as well as some related nonrobotic models. The analysis suggests that, first, research in both cognitive robotics implementations and nonrobotic models needs to be extended to the study of how multiple exteroceptive modalities can be integrated into prediction error minimization schemes. Second, a relevant distinction found here is that cognitive robotics implementations tend to emphasize the learning of a generative model, while in nonrobotics models, it is almost absent. Third, despite the relevance for active inference, few cognitive robotics implementations examine the issues around control and whether it should result from the substitution of inverse models with proprioceptive predictions. Finally, limited attention has been placed on precision weighting and the tracking of prediction error dynamics. These mechanisms should help to explore more complex behaviors and tasks in cognitive robotics research under the predictive processing framework.

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58. On the Achievability of Blind Source Separation for High-Dimensional Nonlinear Source Mixtures

Takuya Isomura, Taro Toyoizumi

https://doi.org/10.1162/neco_a_01378 | pp. 1433-1468

For many years, a combination of principal component analysis (PCA) and independent component analysis (ICA) has been used for blind source separation (BSS). However, it remains unclear why these linear methods work well with real-world data that involve nonlinear source mixtures. This work theoretically validates that a cascade of linear PCA and ICA can solve a nonlinear BSS problem accurately-when the sensory inputs are generated from hidden sources via nonlinear mappings with sufficient dimensionality. Our proposed theorem, termed the asymptotic linearization theorem, theoretically guarantees that applying linear PCA to the inputs can reliably extract a subspace spanned by the linear projections from every hidden source as the major components-and thus projecting the inputs onto their major eigenspace can effectively recover a linear transformation of the hidden sources. Then subsequent application of linear ICA can separate all the true independent hidden sources accurately. Zero-element-wise-error nonlinear BSS is asymptotically attained when the source dimensionality is large and the input dimensionality is sufficiently larger than the source dimensionality. Our proposed theorem is validated analytically and numerically. Moreover, the same computation can be performed by using Hebbian-like plasticity rules, implying the biological plausibility of this nonlinear BSS strategy. Our results highlight the utility of linear PCA and ICA for accurately and reliably recovering nonlinearly mixed sources and suggest the importance of employing sensors with sufficient dimensionality to identify true hidden sources of real-world data.

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59. Critical Point-Finding Methods Reveal Gradient-Flat Regions of Deep Network Losses

Charles G. Frye, James B. Simon, Neha S. Wadia, Andrew Ligeralde, Michael R. DeWeese, Kristofer E. Bouchard

https://doi.org/10.1162/neco_a_01388 | pp. 1469-1497

Despite the fact that the loss functions of deep neural networks are highly nonconvex, gradient-based optimization algorithms converge to approximately the same performance from many random initial points. One thread of work has focused on explaining this phenomenon by numerically characterizing the local curvature near critical points of the loss function, where the gradients are near zero. Such studies have reported that neural network losses enjoy a no-bad-local-minima property, in disagreement with more recent theoretical results. We report here that the methods used to find these putative critical points suffer from a bad local minima problem of their own: they often converge to or pass through regions where the gradient norm has a stationary point. We call these gradient-flat regions, since they arise when the gradient is approximately in the kernel of the Hessian, such that the loss is locally approximately linear, or flat, in the direction of the gradient. We describe how the presence of these regions necessitates care in both interpreting past results that claimed to find critical points of neural network losses and in designing second-order methods for optimizing neural networks.

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60. Reinforcement Learning in Sparse-Reward Environments With Hindsight Policy Gradients

Paulo Rauber, Avinash Ummadisingu, Filipe Mutz, Jürgen Schmidhuber

https://doi.org/10.1162/neco_a_01387 | pp. 1498-1553

A reinforcement learning agent that needs to pursue different goals across episodes requires a goal-conditional policy. In addition to their potential to generalize desirable behavior to unseen goals, such

policies may also enable higher-level planning based on subgoals. In sparse-reward environments, the capacity to exploit information about the degree to which an arbitrary goal has been achieved while another goal was intended appears crucial to enabling sample efficient learning. However, reinforcement learning agents have only recently been endowed with such capacity for hindsight. In this letter, we demonstrate how hindsight can be introduced to policy gradient methods, generalizing this idea to a broad class of successful algorithms. Our experiments on a diverse selection of sparse-reward environments show that hindsight leads to a remarkable increase in sample efficiency.

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61. Might a Single Neuron Solve Interesting Machine Learning Problems Through Successive Computations on Its Dendritic Tree?

Ilenna Jones, Konrad P. Körding

https://doi.org/10.1162/neco_a_01390 | pp. 1554-1571

Physiological experiments have highlighted how the dendrites of biological neurons can nonlinearly process distributed synaptic inputs. However, it is unclear how aspects of a dendritic tree, such as its branched morphology or its repetition of presynaptic inputs, determine neural computation beyond this apparent nonlinearity. Here we use a simple model where the dendrite is implemented as a sequence of thresholded linear units. We manipulate the architecture of this model to investigate the impacts of binary branching constraints and repetition of synaptic inputs on neural computation. We find that models with such manipulations can perform well on machine learning tasks, such as Fashion MNIST or Extended MNIST. We find that model performance on these tasks is limited by binary tree branching and dendritic asymmetry and is improved by the repetition of synaptic inputs to different dendritic branches. These computational experiments further neuroscience theory on how different dendritic properties might determine neural computation of clearly defined tasks.

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62. Shaping Dynamics With Multiple Populations in Low-Rank Recurrent Networks

Manuel Beiran, Alexis Dubreuil, Adrian Valente, Francesca Mastrogiuseppe, Srdjan Ostojic

https://doi.org/10.1162/neco_a_01381 | pp. 1572-1615

An emerging paradigm proposes that neural computations can be understood at the level of dynamic systems that govern low-dimensional trajectories of collective neural activity. How the connectivity structure of a network determines the emergent dynamical system, however, remains to be clarified. Here we consider a novel class of models, gaussian-mixture, low-rank recurrent networks in which the rank of the connectivity matrix and the number of statistically defined populations are independent hyperparameters. We show that the resulting collective dynamics form a dynamical system, where the rank sets the dimensionality and the population structure shapes the dynamics. In particular, the collective dynamics can be described in terms of a simplified effective circuit of interacting latent variables. While having a single global population strongly restricts the possible dynamics, we demonstrate that if the number of populations is large enough, a rank R network can approximate any R-dimensional dynamical system.

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63. Online Mental Fatigue Monitoring via Indirect Brain Dynamics Evaluation

Yuangang Pan, Ivor W. Tsang, Yueming Lyu, Avinash Kumar Singh, Chin■Teng Lin

https://doi.org/10.1162/neco_a_01382 | pp. 1616-1655

Driver mental fatigue leads to thousands of traffic accidents. The increasing quality and availability of low-cost electroencephalogram (EEG) systems offer possibilities for practical fatigue monitoring. However, non-data-driven methods, designed for practical, complex situations, usually rely on handcrafted data statistics of EEG signals. To reduce human involvement, we introduce a data-driven methodology for online mental fatigue detection: self-weight ordinal regression (SWORE). Reaction time (RT), referring to the length of time people take to react to an emergency, is widely considered an objective behavioral measure for mental fatigue state. Since regression methods are sensitive to extreme RTs, we propose an indirect RT estimation based on preferences to explore the relationship between EEG and RT, which generalizes to any scenario when an objective fatigue indicator is available. In particular, SWORE evaluates the noisy EEG signals from multiple channels in terms of two states: shaking state and steady state. Modeling the shaking state can discriminate the reliable channels from the uninformative ones, while modeling the steady state can suppress the task-nonrelevant fluctuation within each channel. In addition, an online generalized Bayesian moment matching (online GBMM) algorithm is proposed to online-calibrate SWORE efficiently per participant. Experimental results with 40 participants show that SWORE can maximally achieve consistent with RT, demonstrating the feasibility and adaptability of our proposed framework in practical mental fatigue estimation.

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64. A Framework of Learning Through Empirical Gain Maximization

Yunlong Feng, Qiang Wu

https://doi.org/10.1162/neco_a_01384 | pp. 1656-1697

We develop in this letter a framework of empirical gain maximization (EGM) to address the robust regression problem where heavy-tailed noise or outliers may be present in the response variable. The idea of EGM is to approximate the density function of the noise distribution instead of approximating the truth function directly as usual. Unlike the classical maximum likelihood estimation that encourages equal importance of all observations and could be problematic in the presence of abnormal observations, EGM schemes can be interpreted from a minimum distance estimation viewpoint and allow the ignorance of those observations. Furthermore, we show that several well-known robust nonconvex regression paradigms, such as Tukey regression and truncated least square regression, can be reformulated into this new framework. We then develop a learning theory for EGM by means of which a unified analysis can be conducted for these well-established but not fully understood regression approaches. This new framework leads to a novel interpretation of existing bounded nonconvex loss functions. Within this new framework, the two seemingly irrelevant terminologies, the well-known Tukey's biweight loss for robust regression and the triweight kernel for nonparametric smoothing, are closely related. More precisely, we show that Tukey's biweight loss can be derived from the triweight kernel. Other frequently employed bounded nonconvex loss functions in machine learning, such as the truncated square loss, the Geman-McClure loss, and the exponential squared loss, can also be reformulated from certain smoothing kernels in statistics. In addition, the new framework enables us to devise new bounded nonconvex loss functions for robust learning.

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65. Noise Robust Projection Rule for Klein Hopfield Neural Networks

Masaki Kobayashi

https://doi.org/10.1162/neco_a_01385 | pp. 1698-1716

Multistate Hopfield models, such as complex-valued Hopfield neural networks (CHNNs), have been used as multistate neural associative memories. Quaternion-valued Hopfield neural networks (QHNNs) reduce the number of weight parameters of CHNNs. The CHNNs and QHNNs have weak noise tolerance by the inherent property of rotational invariance. Klein Hopfield neural networks (KHNNs) improve the noise tolerance by resolving rotational invariance. However, the KHNNs have another disadvantage of self-feedback, a major factor of deterioration in noise tolerance. In this work, the stability conditions of KHNNs are extended. Moreover, the projection rule for KHNNs is modified using the extended conditions. The proposed projection rule improves the noise tolerance by a reduction in self-feedback. Computer simulations support that the proposed projection rule improves the noise tolerance of KHNNs.

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66. Erratum to “A Mean-Field Description of Bursting Dynamics in Spiking Neural Networks with Short-Term Adaptation” by Richard Gast, Helmut Schmidt, and Thomas R. Knösche (Neural Computation , September 2020, Vol. 32, No. 9, pp. 1615–1634)

(authors unknown)

https://doi.org/10.1162/neco_c_01397 | pp. 1717-1717

To implement synaptic depression in our model, we introduced an adaptation variable A to our model, governed by equations 3.2 to 3.4. These equations represent synaptic depression as a convolution with an alpha kernel with rate α and timescale τA. We solved the convolution integral to express it as a set of differential equations (see equations 3.5 and 3.6). However, equation 3.6 contains an erroneous scaling of the input firing rate r with the capitation time constant τA. The convolution integral in equation 3.2 can be expressed as a second-order differential equation: (7.1)$\tau A\ddot{A}$=-2A■-AτA+αr.The equation can be transformed into a set of two coupled first-order differential equations under a simple change of variables: (7.2)τAA■=B,(7.3)τAB■=-2B-A+$\alpha\tau A$r.Equation 7.3 differs from equation 3.6 merely by a constant scaling of r with τA. Since we have never varied τA in this study, this has no impact on our results. Still, the reported values of α for which we find synchronized bursting hold only for synaptic depression given by equations 7.2 and 7.3.

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67. Nonlinear Decoding of Natural Images From Large-Scale Primate Retinal Ganglion Recordings

Young Joon Kim, Nora Brackbill, Eleanor Batty, Jin-Hyung Lee, Catalin Mitelut, William L. Tong, E. J. Chichilnisky, Liam Paninski

https://doi.org/10.1162/neco_a_01395 | pp. 1719-1750

Decoding sensory stimuli from neural activity can provide insight into how the nervous system might interpret the physical environment, and facilitates the development of brain-machine interfaces. Nevertheless, the neural decoding problem remains a significant open challenge. Here, we present an efficient nonlinear decoding approach for inferring natural scene stimuli from the spiking activities of retinal

ganglion cells (RGCs). Our approach uses neural networks to improve on existing decoders in both accuracy and scalability. Trained and validated on real retinal spike data from more than 1000 simultaneously recorded macaque RGC units, the decoder demonstrates the necessity of nonlinear computations for accurate decoding of the fine structures of visual stimuli. Specifically, high-pass spatial features of natural images can only be decoded using nonlinear techniques, while low-pass features can be extracted equally well by linear and nonlinear methods. Together, these results advance the state of the art in decoding natural stimuli from large populations of neurons.

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68. From Univariate to Multivariate Coupling Between Continuous Signals and Point Processes: A Mathematical Framework

Shervin Safavi, Nikos K. Logothetis, Michel Besserve

https://doi.org/10.1162/neco_a_01389 | pp. 1751-1817

Time series data sets often contain heterogeneous signals, composed of both continuously changing quantities and discretely occurring events. The coupling between these measurements may provide insights into key underlying mechanisms of the systems under study. To better extract this information, we investigate the asymptotic statistical properties of coupling measures between continuous signals and point processes. We first introduce martingale stochastic integration theory as a mathematical model for a family of statistical quantities that include the phase locking value, a classical coupling measure to characterize complex dynamics. Based on the martingale central limit theorem, we can then derive the asymptotic gaussian distribution of estimates of such coupling measure that can be exploited for statistical testing. Second, based on multivariate extensions of this result and random matrix theory, we establish a principled way to analyze the low-rank coupling between a large number of point processes and continuous signals. For a null hypothesis of no coupling, we establish sufficient conditions for the empirical distribution of squared singular values of the matrix to converge, as the number of measured signals increases, to the well-known Marchenko-Pastur (MP) law, and the largest squared singular value converges to the upper end of the MP support. This justifies a simple thresholding approach to assess the significance of multivariate coupling. Finally, we illustrate with simulations the relevance of our univariate and multivariate results in the context of neural time series, addressing how to reliably quantify the interplay between multichannel local field potential signals and the spiking activity of a large population of neurons.

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69. Lifelong Classification in Open World With Limited Storage Requirements

Wang Bi, Chen Yang, Xuelian Li, Junfu Chen

https://doi.org/10.1162/neco_a_01391 | pp. 1818-1852

This letter focuses on the problem of lifelong classification in the open world, the goal of which is to achieve an endless process of learning. However, incremental data sets (like the streaming data) in the open world, where the new classes may be emerging, are unsuited for classical classification methods. For addressing this problem, existing methods usually retrain the whole observed data sets with the complex computation and the expensive storage cost. This letter attempts to improve the performance of classification in the open world and decomposes the problem into three subproblems: (1) to reject unknown instances, (2) to classify accepted instances, and (3) to cut the cost of learning. Rejecting unknown instances refers to recognize those instances whose classes are unknown according to the learner, which could reduce the computation

of the retraining process and eliminate the storage of historical data sets. We employ outlier detection for rejecting instances and a variant artificial neural network for classifying with fewer weights. Results on several experiments show that the work is effective. Source code can be found at https://github.com/wangbi1988/Lifelong-learning-in-Open-World-Classification.

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70. X-DC: Explainable Deep Clustering Based on Learnable Spectrogram Templates

Chihiro Watanabe, Hirokazu Kameoka

https://doi.org/10.1162/neco_a_01392 | pp. 1853-1885

Deep neural networks (DNNs) have achieved substantial predictive performance in various speech processing tasks. Particularly, it has been shown that a monaural speech separation task can be successfully solved with a DNN-based method called deep clustering (DC), which uses a DNN to describe the process of assigning a continuous vector to each time-frequency (TF) bin and measure how likely each pair of TF bins is to be dominated by the same speaker. In DC, the DNN is trained so that the embedding vectors for the TF bins dominated by the same speaker are forced to get close to each other. One concern regarding DC is that the embedding process described by a DNN has a black-box structure, which is usually very hard to interpret. The potential weakness owing to the noninterpretable black box structure is that it lacks the flexibility of addressing the mismatch between training and test conditions (caused by reverberation, for instance). To overcome this limitation, in this letter, we propose the concept of explainable deep clustering (X-DC), whose network architecture can be interpreted as a process of fitting learnable spectrogram templates to an input spectrogram followed by Wiener filtering. During training, the elements of the spectrogram templates and their activations are constrained to be nonnegative, which facilitates the sparsity of their values and thus improves interpretability. The main advantage of this framework is that it naturally allows us to incorporate a model adaptation mechanism into the network thanks to its physically interpretable structure. We experimentally show that the proposed X-DC enables us to visualize and understand the clues for the model to determine the embedding vectors while achieving speech separation performance comparable to that of the original DC models.

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71. Skip-Connected Self-Recurrent Spiking Neural Networks With Joint Intrinsic Parameter and Synaptic Weight Training

Wenrui Zhang, Peng Li

https://doi.org/10.1162/neco_a_01393 | pp. 1886-1913

As an important class of spiking neural networks (SNNs), recurrent spiking neural networks (RSNNs) possess great computational power and have been widely used for processing sequential data like audio and text. However, most RSNNs suffer from two problems. First, due to the lack of architectural guidance, random recurrent connectivity is often adopted, which does not guarantee good performance. Second, training of RSNNs is in general challenging, bottlenecking achievable model accuracy. To address these problems, we propose a new type of RSNN, skip-connected self-recurrent SNNs (ScSr-SNNs). Recurrence in ScSr-SNNs is introduced by adding self-recurrent connections to spiking neurons. The SNNs with self-recurrent connections can realize recurrent behaviors similar to those of more complex RSNNs, while the error gradients can be more straightforwardly calculated due to the mostly feedforward nature of the

network. The network dynamics is enriched by skip connections between nonadjacent layers. Moreover, we propose a new backpropagation (BP) method, backpropagated intrinsic plasticity (BIP), to boost the performance of ScSr-SNNs further by training intrinsic model parameters. Unlike standard intrinsic plasticity rules that adjust the neuron's intrinsic parameters according to neuronal activity, the proposed BIP method optimizes intrinsic parameters based on the backpropagated error gradient of a well-defined global loss function in addition to synaptic weight training. Based on challenging speech, neuromorphic speech, and neuromorphic image data sets, the proposed ScSr-SNNs can boost performance by up to 2.85% compared with other types of RSNNs trained by state-of-the-art BP methods.

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72. Classification of Autism Spectrum Disorder From EEG-Based Functional Brain Connectivity Analysis

Noura Alotaibi, Koushik Maharatna

https://doi.org/10.1162/neco_a_01394 | pp. 1914-1941

Autism is a psychiatric condition that is typically diagnosed with behavioral assessment methods. Recent years have seen a rise in the number of children with autism. Since this could have serious health and socioeconomic consequences, it is imperative to investigate how to develop strategies for an early diagnosis that might pave the way to an adequate intervention. In this study, the phase-based functional brain connectivity derived from electroencephalogram (EEG) in a machine learning framework was used to classify the children with autism and typical children in an experimentally obtained data set of 12 autism spectrum disorder (ASD) and 12 typical children. Specifically, the functional brain connectivity networks have quantitatively been characterized by graph-theoretic parameters computed from three proposed approaches based on a standard phase-locking value, which were used as the features in a machine learning environment. Our study was successfully classified between two groups with approximately 95.8% accuracy, 100% sensitivity, and 92% specificity through the trial-averaged phase-locking value (PLV) approach and cubic support vector machine (SVM). This work has also shown that significant changes in functional brain connectivity in ASD children have been revealed at theta band using the aggregated graph-theoretic features. Therefore, the findings from this study offer insight into the potential use of functional brain connectivity as a tool for classifying ASD children.

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73. Analysis of EEG Data Using Complex Geometric Structurization

Eddy Kwessi, Lloyd J. Edwards

https://doi.org/10.1162/neco_a_01398 | pp. 1942-1969

Abstract Electroencephalogram (EEG) is a common tool used to understand brain activities. The data are typically obtained by placing electrodes at the surface of the scalp and recording the oscillations of currents passing through the electrodes. These oscillations can sometimes lead to various interpretations, depending on, for example, the subject's health condition, the experiment carried out, the sensitivity of the tools used, or human manipulations. The data obtained over time can be considered a time series. There is evidence in the literature that epilepsy EEG data may be chaotic. Either way, the Embedding Theory in dynamical systems suggests that time series from a complex system could be used to reconstruct its phase space under proper conditions. In this letter, we propose an analysis of epilepsy EEG time series data based on a novel approach dubbed complex geometric structurization. Complex geometric structurization stems from

the construction of strange attractors using Embedding Theory from dynamical systems. The complex geometric structures are themselves obtained using a geometry tool, the α-shapes from shape analysis. Initial analyses show a proof of concept in that these complex structures capture the expected changes brain in lobes under consideration. Further, a deeper analysis suggests that these complex structures can be used as biomarkers for seizure changes.

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74. A Computational Study on Synaptic Plasticity Regulation and Information Processing in Neuron-Astrocyte Networks

Roman Vuillaume, Jhunlyn Lorenzo, S. Binczak, Sabir Jacquir

https://doi.org/10.1162/neco_a_01399 | pp. 1970-1992

Postsynaptic ionotropic receptors critically shape synaptic currents and underpin their activity-dependent plasticity. In recent years, regulation of expression of these receptors by slow inward and outward currents mediated by gliotransmitter release from astrocytes has come under scrutiny as a potentially important mechanism for the regulation of synaptic information transfer. In this study, we consider a model of astrocyte-regulated synapses to investigate this hypothesis at the level of layered networks of interacting neurons and astrocytes. Our simulations hint that gliotransmission sustains the transfer function across layers, although it decorrelates the neuronal activity from the signal pattern. Overall, our results make clear how astrocytes could transform neuronal activity by inducing a lowfrequency modulation of postsynaptic activity.

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75. NetPyNE Implementation and Scaling of the Potjans-Diesmann Cortical Microcircuit Model

Cecilia Romaro, Fernando Araujo Najman, William W. Lytton, Antônio C. Roque, Salvador Durá-Bernal

https://doi.org/10.1162/neco_a_01400 | pp. 1993-2032

The Potjans-Diesmann cortical microcircuit model is a widely used model originally implemented in NEST. Here, we reimplemented the model using NetPyNE, a high-level Python interface to the NEURON simulator, and reproduced the findings of the original publication. We also implemented a method for scaling the network size that preserves first- and second-order statistics, building on existing work on network theory. Our new implementation enabled the use of more detailed neuron models with multicompartmental morphologies and multiple biophysically realistic ion channels. This opens the model to new research, including the study of dendritic processing, the influence of individual channel parameters, the relation to local field potentials, and other multiscale interactions. The scaling method we used provides flexibility to increase or decrease the network size as needed when running these CPU-intensive detailed simulations. Finally, NetPyNE facilitates modifying or extending the model using its declarative language; optimizing model parameters; running efficient, large-scale parallelized simulations; and analyzing the model through built-in methods, including local field potential calculation and information flow measures.

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76. Simulating and Predicting Dynamical Systems With Spatial Semantic Pointers

Aaron R. Voelker, Peter Blouw, Xuan Choo, Nicole Sandra-Yaffa Dumont, Terrence C. Stewart, Chris Eliasmith

https://doi.org/10.1162/neco_a_01410 | pp. 2033-2067

While neural networks are highly effective at learning task-relevant representations from data, they typically do not learn representations with the kind of symbolic structure that is hypothesized to support high-level cognitive processes, nor do they naturally model such structures within problem domains that are continuous in space and time. To fill these gaps, this work exploits a method for defining vector representations that bind discrete (symbol-like) entities to points in continuous topological spaces in order to simulate and predict the behavior of a range of dynamical systems. These vector representations are spatial semantic pointers (SSPs), and we demonstrate that they can (1) be used to model dynamical systems involving multiple objects represented in a symbol-like manner and (2) be integrated with deep neural networks to predict the future of physical trajectories. These results help unify what have traditionally appeared to be disparate approaches in machine learning.

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77. Frequency Selectivity of Neural Circuits With Heterogeneous Discrete Transmission Delays

Akke Mats Houben

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Neurons are connected to other neurons by axons and dendrites that conduct signals with finite velocities, resulting in delays between the firing of a neuron and the arrival of the resultant impulse at other neurons. Since delays greatly complicate the analytical treatment and interpretation of models, they are usually neglected or taken to be uniform, leading to a lack in the comprehension of the effects of delays in neural systems. This letter shows that heterogeneous transmission delays make small groups of neurons respond selectively to inputs with differing frequency spectra. By studying a single integrate-and-fire neuron receiving correlated time-shifted inputs, it is shown how the frequency response is linked to both the strengths and delay times of the afferent connections. The results show that incorporating delays alters the functioning of neural networks, and changes the effect that neural connections and synaptic strengths have.

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78. Learning Brain Dynamics With Coupled Low-Dimensional Nonlinear Oscillators and Deep Recurrent Networks

Germán Abrevaya, Guillaume Dumas, Aleksandr Y. Aravkin, Peng Zheng, Jean-Christophe Gagnon-Audet, James Kozloski, Pablo Polosecki, Guillaume Lajoie, David Cox, Silvina Ponce Dawson, Guillermo Cecchi, Irina Rish

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Many natural systems, especially biological ones, exhibit complex multivariate nonlinear dynamical behaviors that can be hard to capture by linear autoregressive models. On the other hand, generic nonlinear models such as deep recurrent neural networks often require large amounts of training data, not always available in domains such as brain imaging; also, they often lack interpretability. Domain knowledge about the types of dynamics typically observed in such systems, such as a certain type of dynamical systems models, could complement purely data-driven techniques by providing a good prior. In this work, we consider a class of ordinary differential equation (ODE) models known as van der Pol (VDP) oscil lators and

evaluate their ability to capture a low-dimensional representation of neural activity measured by different brain imaging modalities, such as calcium imaging (CaI) and fMRI, in different living organisms: larval zebrafish, rat, and human. We develop a novel and efficient approach to the nontrivial problem of parameters estimation for a network of coupled dynamical systems from multivariate data and demonstrate that the resulting VDP models are both accurate and interpretable, as VDP's coupling matrix reveals anatomically meaningful excitatory and inhibitory interactions across different brain subsystems. VDP outperforms linear autoregressive models (VAR) in terms of both the data fit accuracy and the quality of insight provided by the coupling matrices and often tends to generalize better to unseen data when predicting future brain activity, being comparable to and sometimes better than the recurrent neural networks (LSTMs). Finally, we demonstrate that our (generative) VDP model can also serve as a data-augmentation tool leading to marked improvements in predictive accuracy of recurrent neural networks. Thus, our work contributes to both basic and applied dimensions of neuroimaging: gaining scientific insights and improving brain-based predictive models, an area of potentially high practical importance in clinical diagnosis and neurotechnology.

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79. Direction Matters: On Influence-Preserving Graph Summarization and Max-Cut Principle for Directed Graphs

Wenkai Xu, Gang Niu, Aapo Hyvärinen, Masashi Sugiyama

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Summarizing large-scale directed graphs into small-scale representations is a useful but less-studied problem setting. Conventional clustering approaches, based on Min-Cut-style criteria, compress both the vertices and edges of the graph into the communities, which lead to a loss of directed edge information. On the other hand, compressing the vertices while preserving the directed-edge information provides a way to learn the small-scale representation of a directed graph. The reconstruction error, which measures the edge information preserved by the summarized graph, can be used to learn such representation. Compared to the original graphs, the summarized graphs are easier to analyze and are capable of extracting group-level features, useful for efficient interventions of population behavior. In this letter, we present a model, based on minimizing reconstruction error with nonnegative constraints, which relates to a Max-Cut criterion that simultaneously identifies the compressed nodes and the directed compressed relations between these nodes. A multiplicative update algorithm with column-wise normalization is proposed. We further provide theoretical results on the identifiability of the model and the convergence of the proposed algorithms. Experiments are conducted to demonstrate the accuracy and robustness of the proposed method.

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80. Artificial Neural Variability for Deep Learning: On Overfitting, Noise Memorization, and Catastrophic Forgetting

Zeke Xie, Fengxiang He, Shaopeng Fu, Issei Sato, Dacheng Tao, Masashi Sugiyama

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Deep learning is often criticized by two serious issues that rarely exist in natural nervous systems: overfitting and catastrophic forgetting. It can even memorize randomly labeled data, which has little knowledge behind the instance-label pairs. When a deep network continually learns over time by accommodating new tasks, it usually quickly overwrites the knowledge learned from previous tasks. Referred to as the neural variability, it

is well known in neuroscience that human brain reactions exhibit substantial variability even in response to the same stimulus. This mechanism balances accuracy and plasticity/flexibility in the motor learning of natural nervous systems. Thus, it motivates us to design a similar mechanism, named artificial neural variability (ANV), that helps artificial neural networks learn some advantages from "natural" neural networks. We rigorously prove that ANV plays as an implicit regularizer of the mutual information between the training data and the learned model. This result theoretically guarantees ANV a strictly improved generalizability, robustness to label noise, and robustness to catastrophic forgetting. We then devise a neural variable risk minimization (NVRM) framework and neural variable optimizers to achieve ANV for conventional network architectures in practice. The empirical studies demonstrate that NVRM can effectively relieve overfitting, label noise memorization, and catastrophic forgetting at negligible costs.

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81. Power Function Error Initialization Can Improve Convergence of Backpropagation Learning in Neural Networks for Classification

Andreas Knoblauch

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Supervised learning corresponds to minimizing a loss or cost function expressing the differences between model predictions yn and the target values tn given by the training data. In neural networks, this means backpropagating error signals through the transposed weight matrixes from the output layer toward the input layer. For this, error signals in the output layer are typically initialized by the difference yn- tn, which is optimal for several commonly used loss functions like cross-entropy or sum of squared errors. Here I evaluate a more general error initialization method using power functions |yn- tn|q for q>0, corresponding to a new family of loss functions that generalize cross-entropy. Surprisingly, experiments on various learning tasks reveal that a proper choice of q can significantly improve the speed and convergence of backpropagation learning, in particular in deep and recurrent neural networks. The results suggest two main reasons for the observed improvements. First, compared to cross-entropy, the new loss functions provide better fits to the distribution of error signals in the output layer and therefore maximize the model's likelihood more efficiently. Second, the new error initialization procedure may often provide a better gradient-to-loss ratio over a broad range of neural output activity, thereby avoiding flat loss landscapes with vanishing gradients.

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82. Storage Capacity of Quaternion-Valued Hopfield Neural Networks With Dual Connections

Masaki Kobayashi

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A complex-valued Hopfield neural network (CHNN) is a multistate Hopfield model. A quaternion-valued Hopfield neural network (QHNN) with a twin-multistate activation function was proposed to reduce the number of weight parameters of CHNN. Dual connections (DCs) are introduced to the QHNNs to improve the noise tolerance. The DCs take advantage of the noncommutativity of quaternions and consist of two weights between neurons. A QHNN with DCs provides much better noise tolerance than a CHNN. Although a CHNN and a QHNN with DCs have the samenumber of weight parameters, the storage capacity of projection rule for QHNNs with DCs is half of that for CHNNs and equals that of conventional QHNNs. The

small storage capacity of QHNNs with DCs is caused by projection rule, not the architecture. In this work, the ebbian rule is introduced and proved by stochastic analysis that the storage capacity of a QHNN with DCs is 0.8 times as many as that of a CHNN.

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83. Randomized Self-Organizing Map

Nicolas P. Rougier, Georgios Detorakis
https://doi.org/10.1162/neco_a_01406 | pp. 2241-2273

We propose a variation of the self-organizing map algorithm by considering the random placement of neurons on a two-dimensional manifold, following a blue noise distribution from which various topologies can be derived. These topologies possess random (but controllable) discontinuities that allow for a more flexible self-organization, especially with high-dimensional data. The proposed algorithm is tested on one-, two- and three-dimensional tasks, as well as on the MNIST handwritten digits data set and validated using spectral analysis and topological data analysis tools. We also demonstrate the ability of the randomized self-organizing map to gracefully reorganize itself in case of neural lesion and/or neurogenesis.

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84. Pathological Spectra of the Fisher Information Metric and Its Variants in Deep Neural Networks

Ryo Karakida, Shotaro Akaho, ■■■-■■■ ■■■■■
https://doi.org/10.1162/neco_a_01411 | pp. 2274-2307

The Fisher information matrix (FIM) plays an essential role in statistics and machine learning as a Riemannian metric tensor or a component of the Hessian matrix of loss functions. Focusing on the FIM and its variants in deep neural networks (DNNs), we reveal their characteristic scale dependence on the network width, depth, and sample size when the network has random weights and is sufficiently wide. This study covers two widely used FIMs for regression with linear output and for classification with softmax output. Both FIMs asymptotically show pathological eigenvalue spectra in the sense that a small number of eigenvalues become large outliers depending on the width or sample size, while the others are much smaller. It implies that the local shape of the parameter space or loss landscape is very sharp in a few specific directions while almost flat in the other directions. In particular, the softmax output disperses the outliers and makes a tail of the eigenvalue density spread from the bulk. We also show that pathological spectra appear in other variants of FIMs: one is the neural tangent kernel; another is a metric for the input signal and feature space that arises from feedforward signal propagation. Thus, we provide a unified perspective on the FIM and its variants that will lead to more quantitative understanding of learning in large-scale DNNs.

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85. A Biologically Plausible Neural Network for Multichannel Canonical Correlation Analysis

David Lipshutz, Yanis Bahroun, Siavash Golkar, Anirvan M. Sengupta, Dmitri B. Chklovskii
https://doi.org/10.1162/neco_a_01414 | pp. 2309-2352

Cortical pyramidal neurons receive inputs from multiple distinct neural populations and integrate these inputs in separate dendritic compartments. We explore the possibility that cortical microcircuits implement

canonical correlation analysis (CCA), an unsupervised learning method that projects the inputs onto a common subspace so as to maximize the correlations between the projections. To this end, we seek a multichannel CCA algorithm that can be implemented in a biologically plausible neural network. For biological plausibility, we require that the network operates in the online setting and its synaptic update rules are local. Starting from a novel CCA objective function, we derive an online optimization algorithm whose optimization steps can be implemented in a single-layer neural network with multicompartmental neurons and local non-Hebbian learning rules. We also derive an extension of our online CCA algorithm with adaptive output rank and output whitening. Interestingly, the extension maps onto a neural network whose neural architecture and synaptic updates resemble neural circuitry and non-Hebbian plasticity observed in the cortex.

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86. Emergence of Content-Agnostic Information Processing by a Robot Using Active Inference, Visual Attention, Working Memory, and Planning

Jeffrey Queißer, Minju Jung, Takazumi Matsumoto, Jun Tani

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Generalization by learning is an essential cognitive competency for humans. For example, we can manipulate even unfamiliar objects and can generate mental images before enacting a preplan. How is this possible? Our study investigated this problem by revisiting our previous study (Jung, Matsumoto, & Tani, 2019), which examined the problem of vision-based, goal-directed planning by robots performing a task of block stacking. By extending the previous study, our work introduces a large network comprising dynamically interacting submodules, including visual working memory (VWMs), a visual attention module, and an executive network. The executive network predicts motor signals, visual images, and various controls for attention, as well as masking of visual information. The most significant difference from the previous study is that our current model contains an additional VWM. The entire network is trained by using predictive coding and an optimal visuomotor plan to achieve a given goal state is inferred using active inference. Results indicate that our current model performs significantly better than that used in Jung et al. (2019), especially when manipulating blocks with unlearned colors and textures. Simulation results revealed that the observed generalization was achieved because content-agnostic information processing developed through synergistic interaction between the second VWM and other modules during the course of learning, in which memorizing image contents and transforming them are dissociated. This letter verifies this claim by conducting both qualitative and quantitative analysis of simulation results.

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87. Bridging M/EEG Source Imaging and Independent Component Analysis Frameworks Using Biologically Inspired Sparsity Priors

Alejandro Ojeda, Kenneth Kreutz-Delgado, Jyoti Mishra

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Electromagnetic source imaging (ESI) and independent component analysis (ICA) are two popular and apparently dissimilar frameworks for M/EEG analysis. This letter shows that the two frameworks can be linked by choosing biologically inspired source sparsity priors. We demonstrate that ESI carried out by the sparse Bayesian learning (SBL) algorithm yields source configurations composed of a few active regions that are also maximally independent from one another. In addition, we extend the standard SBL approach to

source imaging in two important directions. First, we augment the generative model of M/EEG to include artifactual sources. Second, we modify SBL to allow for efficient model inversion with sequential data. We refer to this new algorithm as recursive SBL (RSBL), a source estimation filter with potential for online and offline imaging applications. We use simulated data to verify that RSBL can accurately estimate and demix cortical and artifactual sources under different noise conditions. Finally, we show that on real error-related EEG data, RSBL can yield single-trial source estimates in agreement with the experimental literature. Overall, by demonstrating that ESI can produce maximally independent sources while simultaneously localizing them in cortical space, we bridge the gap between the ESI and ICA frameworks for M/EEG analysis.

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88. Few-Shot Learning in Spiking Neural Networks by Multi-Timescale Optimization

Runhao Jiang, Jie Zhang, Rui Yan, Huajin Tang

https://doi.org/10.1162/neco_a_01423 | pp. 2439-2472

Learning new concepts rapidly from a few examples is an open issue in spike-based machine learning. This few-shot learning imposes substantial challenges to the current learning methodologies of spiking neuron networks (SNNs) due to the lack of task-related priori knowledge. The recent learning-to-learn (L2L) approach allows SNNs to acquire priori knowledge through example-level learning and task-level optimization. However, existing L2L-based frameworks do not target the neural dynamics (i.e., neuronal and synaptic parameter changes) on different timescales. This diversity of temporal dynamics is an important attribute in spike-based learning, which facilitates the networks to rapidly acquire knowledge from very few examples and gradually integrate this knowledge. In this work, we consider the neural dynamics on various timescales and provide a multi-timescale optimization (MTSO) framework for SNNs. This framework introduces an adaptive-gated LSTM to accommodate two different timescales of neural dynamics: short-term learning and long-term evolution. Short-term learning is a fast knowledge acquisition process achieved by a novel surrogate gradient online learning (SGOL) algorithm, where the LSTM guides gradient updating of SNN on a short timescale through an adaptive learning rate and weight decay gating. The long-term evolution aims to slowly integrate acquired knowledge and form a priori, which can be achieved by optimizing the LSTM guidance process to tune SNN parameters on a long timescale. Experimental results demonstrate that the collaborative optimization of multi-timescale neural dynamics can make SNNs achieve promising performance for the few-shot learning tasks.

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89. Least kth-Order and Rényi Generative Adversarial Networks

Himesh Bhatia, William Paul, Fady Alajaji, Bahman Gharesifard, Philippe Burlina

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We investigate the use of parameterized families of information-theoretic measures to generalize the loss functions of generative adversarial networks (GANs) with the objective of improving performance. A new generator loss function, least kth-order GAN (LkGAN), is introduced, generalizing the least squares GANs (LSGANs) by using a kth-order absolute error distortion measure with $k \geq 1$ (which recovers the LSGAN loss function when $k=2$). It is shown that minimizing this generalized loss function under an (unconstrained) optimal discriminator is equivalent to minimizing the kth-order Pearson-Vajda divergence. Another novel

GAN generator loss function is next proposed in terms of Rényi cross-entropy functionals with order $\alpha>0$, $\alpha\neq1$. It is demonstrated that this Rényi-centric generalized loss function, which provably reduces to the original GAN loss function as $\alpha\rightarrow1$, preserves the equilibrium point satisfied by the original GAN based on the Jensen-Rényi divergence, a natural extension of the Jensen-Shannon divergence. Experimental results indicate that the proposed loss functions, applied to the MNIST and CelebA data sets, under both DCGAN and StyleGAN architectures, confer performance benefits by virtue of the extra degrees of freedom provided by the parameters k and α, respectively. More specifically, experiments show improvements with regard to the quality of the generated images as measured by the Fréchet inception distance score and training stability. While it was applied to GANs in this study, the proposed approach is generic and can be used in other applications of information theory to deep learning, for example, the issues of fairness or privacy in artificial intelligence.

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90. Do Neural Networks for Segmentation Understand Insideness?

Kimberly Villalobos, Vilim Štih, Amineh Ahmadinejad, Shobhita Sundaram, Jamell Dozier, Andrew Francl, Frederico Luis de Azevedo, Tomotake Sasaki, Xavier Boix

https://doi.org/10.1162/neco_a_01413 | pp. 2511-2549

The insideness problem is an aspect of image segmentation that consists of determining which pixels are inside and outside a region. Deep neural networks (DNNs) excel in segmentation benchmarks, but it is unclear if they have the ability to solve the insideness problem as it requires evaluating long-range spatial dependencies. In this letter, we analyze the insideness problem in isolation, without texture or semantic cues, such that other aspects of segmentation do not interfere in the analysis. We demonstrate that DNNs for segmentation with few units have sufficient complexity to solve the insideness for any curve. Yet such DNNs have severe problems with learning general solutions. Only recurrent networks trained with small images learn solutions that generalize well to almost any curve. Recurrent networks can decompose the evaluation of long-range dependencies into a sequence of local operations, and learning with small images alleviates the common difficulties of training recurrent networks with a large number of unrolling steps.

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91. On Neural Associative Memory Structures: Storage and Retrieval of Sequences in a Chain of Tournaments

Asieh Abolpour Mofrad, Samaneh Abolpour Mofrad, Anis Yazidi, Matthew G. Parker

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Associative memories enjoy many interesting properties in terms of error correction capabilities, robustness to noise, storage capacity, and retrieval performance, and their usage spans over a large set of applications. In this letter, we investigate and extend tournament-based neural networks, originally proposed by Jiang, Gripon, Berrou, and Rabbat (2016), a novel sequence storage associative memory architecture with high memory efficiency and accurate sequence retrieval. We propose a more general method for learning the sequences, which we call feedback tournament-based neural networks. The retrieval process is also extended to both directions: forward and backward-in other words, any large-enough segment of a sequence can produce the whole sequence. Furthermore, two retrieval algorithms, cache-winner and explore-winner, are introduced to increase the retrieval performance. Through simulation results, we shed light on the strengths and weaknesses of each algorithm.

92. Statistical Properties of Color Matching Functions

María da Fonseca, Inés Samengo

https://doi.org/10.1162/neco_a_01421 | pp. 2578-2601

In trichromats, color vision entails the projection of an infinite-dimensional space (the one containing all possible electromagnetic power spectra) onto the three-dimensional space that modulates the activity of the three types of cones. This drastic reduction in dimensionality gives rise to metamerism, that is, the perceptual chromatic equivalence between two different light spectra. The classes of equivalence of metamerism are revealed by color-matching experiments in which observers adjust the intensity of three monochromatic light beams of three preset wavelengths (the primaries) to produce a mixture that is perceptually equal to a given monochromatic target stimulus. Here we use the linear relation between the color matching functions and the absorption probabilities of each type of cone to find particularly useful triplets of primaries. As a second goal, we also derive an analytical description of the trial-to-trial variability and the correlations of color matching functions stemming from Poissonian noise in photon capture. We analyze how the statistical properties of the responses to color-matching experiments vary with the retinal composition and the wavelengths of peak absorption probability, and compare them with experimental data on subject-to-subject variability obtained previously.

93. Stimulus-Driven and Spontaneous Dynamics in Excitatory-Inhibitory Recurrent Neural Networks for Sequence Representation

Alfred Ajay Aureate Rajakumar, John Rinzel, Zhe Chen

https://doi.org/10.1162/neco_a_01418 | pp. 2603-2645

Recurrent neural networks (RNNs) have been widely used to model sequential neural dynamics ("neural sequences") of cortical circuits in cognitive and motor tasks. Efforts to incorporate biological constraints and Dale's principle will help elucidate the neural representations and mechanisms of underlying circuits. We trained an excitatory-inhibitory RNN to learn neural sequences in a supervised manner and studied the representations and dynamic attractors of the trained network. The trained RNN was robust to trigger the sequence in response to various input signals and interpolated a time-warped input for sequence representation. Interestingly, a learned sequence can repeat periodically when the RNN evolved beyond the duration of a single sequence. The eigenspectrum of the learned recurrent connectivity matrix with growing or damping modes, together with the RNN's nonlinearity, were adequate to generate a limit cycle attractor. We further examined the stability of dynamic attractors while training the RNN to learn two sequences. Together, our results provide a general framework for understanding neural sequence representation in the excitatory-inhibitory RNN.

94. Restricted Boltzmann Machines as Models of Interacting Variables

Nicola Bulso, Yasser Roudi

https://doi.org/10.1162/neco_a_01420 | pp. 2646-2681

We study the type of distributions that restricted Boltzmann machines (RBMs) with different activation functions can express by investigating the effect of the activation function of the hidden nodes on the

marginal distribution they impose on observed binary nodes. We report an exact expression for these marginals in the form of a model of interacting binary variables with the explicit form of the interactions depending on the hidden node activation function. We study the properties of these interactions in detail and evaluate how the accuracy with which the RBM approximates distributions over binary variables depends on the hidden node activation function and the number of hidden nodes. When the inferred RBM parameters are weak, an intuitive pattern is found for the expression of the interaction terms, which reduces substantially the differences across activation functions. We show that the weak parameter approximation is a good approximation for different RBMs trained on the MNIST data set. Interestingly, in these cases, the mapping reveals that the inferred models are essentially low order interaction models.

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95. Chance-Constrained Active Inference

Thijs van de Laar, ■smai■l ■enöz, Ayça Özçelikkale, Henk Wymeersch

https://doi.org/10.1162/neco_a_01427 | pp. 2710-2735

Active inference (ActInf) is an emerging theory that explains perception and action in biological agents in terms of minimizing a free energy bound on Bayesian surprise. Goal-directed behavior is elicited by introducing prior beliefs on the underlying generative model. In contrast to prior beliefs, which constrain all realizations of a random variable, we propose an alternative approach through chance constraints, which allow for a (typically small) probability of constraint violation, and demonstrate how such constraints can be used as intrinsic drivers for goal-directed behavior in ActInf. We illustrate how chance-constrained ActInf weights all imposed (prior) constraints on the generative model, allowing, for example, for a trade-off between robust control and empirical chance constraint violation. Second, we interpret the proposed solution within a message passing framework. Interestingly, the message passing interpretation is not only relevant to the context of ActInf, but also provides a general-purpose approach that can account for chance constraints on graphical models. The chance constraint message updates can then be readily combined with other prederived message update rules without the need for custom derivations. The proposed chance-constrained message passing framework thus accelerates the search for workable models in general and can be used to complement message-passing formulations on generative neural models.

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96. Multibranch Formal Neuron: An Internally Nonlinear Learning Unit

Marifi Güler

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The transformation of synaptic input into action potential in nerve cells is strongly influenced by the morphology of the dendritic arbor as well as the synaptic efficacy map. The multiplicity of dendritic branches strikingly enables a single cell to act as a highly nonlinear processing element. Studies have also found functional synaptic clustering whereby synapses that encode a common sensory feature are spatially clustered together on the branches. Motivated by these findings, here we introduce a multibranch formal model of the neuron that can integrate synaptic inputs nonlinearly through collective action of its dendritic branches and yields synaptic clustering. An analysis in support of its use as a computational building block is offered. Also offered is an accompanying gradient descent-based learning algorithm. The model unit spans a wide spectrum of nonlinearities, including the parity problem, and can outperform the multilayer perceptron in generalizing to unseen data. The occurrence of synaptic clustering boosts the generalization

efficiency of the unit, which may also be the answer for the puzzling ubiquity of synaptic clustering in the real neurons. Our theoretical analysis is backed up by simulations. The study could pave the way to new artificial neural networks.

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97. Realizing Active Inference in Variational Message Passing: The Outcome-Blind Certainty Seeker

Théophile Champion, Marek Grze■, Howard Bowman

https://doi.org/10.1162/neco_a_01422 | pp. 2762-2826

Active inference is a state-of-the-art framework in neuroscience that offers a unified theory of brain function. It is also proposed as a framework for planning in AI. Unfortunately, the complex mathematics required to create new models can impede application of active inference in neuroscience and AI research. This letter addresses this problem by providing a complete mathematical treatment of the active inference framework in discrete time and state spaces and the derivation of the update equations for any new model. We leverage the theoretical connection between active inference and variational message passing as described by John Winn and Christopher M. Bishop in 2005. Since variational message passing is a well-defined methodology for deriving Bayesian belief update equations, this letter opens the door to advanced generative models for active inference. We show that using a fully factorized variational distribution simplifies the expected free energy, which furnishes priors over policies so that agents seek unambiguous states. Finally, we consider future extensions that support deep tree searches for sequential policy optimization based on structure learning and belief propagation.

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98. Integration of Leaky-Integrate-and-Fire Neurons in Standard Machine Learning Architectures to Generate Hybrid Networks: A Surrogate Gradient Approach

Richard Gerum, Achim Schilling

https://doi.org/10.1162/neco_a_01424 | pp. 2827-2852

Up to now, modern machine learning (ML) has been based on approximating big data sets with high-dimensional functions, taking advantage of huge computational resources. We show that biologically inspired neuron models such as the leaky-integrate-and-fire (LIF) neuron provide novel and efficient ways of information processing. They can be integrated in machine learning models and are a potential target to improve ML performance. Thus, we have derived simple update rules for LIF units to numerically integrate the differential equations. We apply a surrogate gradient approach to train the LIF units via backpropagation. We demonstrate that tuning the leak term of the LIF neurons can be used to run the neurons in different operating modes, such as simple signal integrators or coincidence detectors. Furthermore, we show that the constant surrogate gradient, in combination with tuning the leak term of the LIF units, can be used to achieve the learning dynamics of more complex surrogate gradients. To prove the validity of our method, we applied it to established image data sets (the Oxford 102 flower data set, MNIST), implemented various network architectures, used several input data encodings and demonstrated that the method is suitable to achieve state-of-the-art classification performance. We provide our method as well as further surrogate gradient methods to train spiking neural networks via backpropagation as an open-source KERAS package to make it available to the neuroscience and machine learning community. To increase the interpretability of

the underlying effects and thus make a small step toward opening the black box of machine learning, we provide interactive illustrations, with the possibility of systematically monitoring the effects of parameter changes on the learning characteristics.

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99. Multilinear Common Component Analysis via Kronecker Product Representation

Kohei Yoshikawa, Shuichi Kawano

https://doi.org/10.1162/neco_a_01425 | pp. 2853-2880

We consider the problem of extracting a common structure from multiple tensor data sets. For this purpose, we propose multilinear common component analysis (MCCA) based on Kronecker products of mode-wise covariance matrices. MCCA constructs a common basis represented by linear combinations of the original variables that lose little information of the multiple tensor data sets. We also develop an estimation algorithm for MCCA that guarantees mode-wise global convergence. Numerical studies are conducted to show the effectiveness of MCCA.

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100. A Correspondence Between Normalization Strategies in Artificial and Biological Neural Networks

Yang Shen, Julia Wang, Saket Navlakha

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A fundamental challenge at the interface of machine learning and neuroscience is to uncover computational principles that are shared between artificial and biological neural networks. In deep learning, normalization methods such as batch normalization, weight normalization, and their many variants help to stabilize hidden unit activity and accelerate network training, and these methods have been called one of the most important recent innovations for optimizing deep networks. In the brain, homeostatic plasticity represents a set of mechanisms that also stabilize and normalize network activity to lie within certain ranges, and these mechanisms are critical for maintaining normal brain function. In this article, we discuss parallels between artificial and biological normalization methods at four spatial scales: normalization of a single neuron's activity, normalization of synaptic weights of a neuron, normalization of a layer of neurons, and normalization of a network of neurons. We argue that both types of methods are functionally equivalent-that is, both push activation patterns of hidden units toward a homeostatic state, where all neurons are equally used-and we argue that such representations can improve coding capacity, discrimination, and regularization. As a proof of concept, we develop an algorithm, inspired by a neural normalization technique called synaptic scaling, and show that this algorithm performs competitively against existing normalization methods on several data sets. Overall, we hope this bidirectional connection will inspire neuroscientists and machine learners in three ways: to uncover new normalization algorithms based on established neurobiological principles; to help quantify the trade-offs of different homeostatic plasticity mechanisms used in the brain; and to offer insights about how stability may not hinder, but may actually promote, plasticity.

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101. On PDE Characterization of Smooth Hierarchical Functions Computed by Neural Networks

Khashayar Filom, Roozbeh Farhoodi, Konrad P. Körding

https://doi.org/10.1162/neco_a_01441 | pp. 3204-3263

Neural networks are versatile tools for computation, having the ability to approximate a broad range of functions. An important problem in the theory of deep neural networks is expressivity; that is, we want to understand the functions that are computable by a given network. We study real, infinitely differentiable (smooth) hierarchical functions implemented by feedforward neural networks via composing simpler functions in two cases: (1) each constituent function of the composition has fewer inputs than the resulting function and (2) constituent functions are in the more specific yet prevalent form of a nonlinear univariate function (e.g., tanh) applied to a linear multivariate function. We establish that in each of these regimes, there exist nontrivial algebraic partial differential equations (PDEs) that are satisfied by the computed functions. These PDEs are purely in terms of the partial derivatives and are dependent only on the topology of the network. Conversely, we conjecture that such PDE constraints, once accompanied by appropriate nonsingularity conditions and perhaps certain inequalities involving partial derivatives, guarantee that the smooth function under consideration can be represented by the network. The conjecture is verified in numerous examples, including the case of tree architectures, which are of neuroscientific interest. Our approach is a step toward formulating an algebraic description of functional spaces associated with specific neural networks, and may provide useful new tools for constructing neural networks.

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102. Learning the Synaptic and Intrinsic Membrane Dynamics Underlying Working Memory in Spiking Neural Network Models

Yinghao Li, Robert Kim, Terrence J. Sejnowski

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Recurrent neural network (RNN) models trained to perform cognitive tasks are a useful computational tool for understanding how cortical circuits execute complex computations. However, these models are often composed of units that interact with one another using continuous signals and overlook parameters intrinsic to spiking neurons. Here, we developed a method to directly train not only synaptic-related variables but also membrane-related parameters of a spiking RNN model. Training our model on a wide range of cognitive tasks resulted in diverse yet task-specific synaptic and membrane parameters. We also show that fast membrane time constants and slow synaptic decay dynamics naturally emerge from our model when it is trained on tasks associated with working memory (WM). Further dissecting the optimized parameters revealed that fast membrane properties are important for encoding stimuli, and slow synaptic dynamics are needed for WM maintenance. This approach offers a unique window into how connectivity patterns and intrinsic neuronal properties contribute to complex dynamics in neural populations.

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103. Burster Reconstruction Considering Unmeasurable Variables in the Epileptor Model

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Epilepsy is one of the most common brain disorders worldwide, affecting millions of people every year. Although significant effort has been put into better understanding it and mitigating its effects, the conventional treatments are not fully effective. Advances in computational neuroscience, using mathematical dynamic models that represent brain activities at different scales, have enabled addressing epilepsy from a more theoretical standpoint. In particular, the recently proposed Epileptor model stands out among these models, because it represents well the main features of seizures, and the results from its simulations have been consistent with experimental observations. In addition, there has been an increasing interest in designing control techniques for Epileptor that might lead to possible realistic feedback controllers in the future. However, such approaches rely on knowing all of the states of the model, which is not the case in practice. The work explored in this letter aims to develop a state observer to estimate Epileptor's unmeasurable variables, as well as reconstruct the respective so-called bursters. Furthermore, an alternative modeling is presented for enhancing the convergence speed of an observer. The results show that the proposed approach is efficient under two main conditions: when the brain is undergoing a seizure and when a transition from the healthy to the epileptiform activity occurs.

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104. Simple Convolutional-Based Models: Are They Learning the Task or the Data?

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Convolutional neural networks (CNNs) evolved from Fukushima's neocognitron model, which is based on the ideas of Hubel and Wiesel about the early stages of the visual cortex. Unlike other branches of neocognitron-based models, the typical CNN is based on end-to-end supervised learning by backpropagation and removes the focus from built-in invariance mechanisms, using pooling not as a way to tolerate small shifts but as a regularization tool that decreases model complexity. These properties of end-to-end supervision and flexibility of structure allow the typical CNN to become highly tuned to the training data, leading to extremely high accuracies on typical visual pattern recognition data sets. However, in this work, we hypothesize that there is a flip side to this capability, a hidden overfitting. More concretely, a supervised, backpropagation based CNN will outperform a neocognitron/map transformation cascade (MTC) when trained and tested inside the same data set. Yet if we take both models trained and test them on the same task but on another data set (without retraining), the overfitting appears. Other neocognitron descendants like the What-Where model go in a different direction. In these models, learning remains unsupervised, but more structure is added to capture invariance to typical changes. Knowing that, we further hypothesize that if we repeat the same experiments with this model, the lack of supervision may make it worse than the typical CNN inside the same data set, but the added structure will make it generalize even better to another one. To put our hypothesis to the test, we choose the simple task of handwritten digit classification and take two well-known data sets of it: MNIST and ETL-1. To try to make the two data sets as similar as possible, we experiment with several types of preprocessing. However, regardless of the type in question, the results align exactly with expectation.

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105. Asymmetric Weights and Retrieval Practice in an Autoassociative Neural Network Model of Paired-Associate Learning

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Rizzuto and Kahana (2001) applied an autoassociative Hopfield network to a paired-associate word learning experiment in which (1) participants studied word pairs (e.g., ABSENCE-HOLLOW), (2) were tested in one direction (ABSENCE-?) on a first test, and (3) were tested in the same direction again or in the reverse direction (?-HOLLOW) on a second test. The model contained a correlation parameter to capture the dependence between forward versus backward learning between the two words of a word pair, revealing correlation values close to 1.0 for all participants, consistent with neural network models that use the same weight for communication in both directions between nodes. We addressed several limitations of the model simulations and proposed two new models incorporating retrieval practice learning (e.g., the effect of the first test on the second) that fit the accuracy data more effectively, revealing substantially lower correlation values (average of .45 across participants, with zero correlation for some participants). In addition, we analyzed recall latencies, finding that second test recall was faster in the same direction after a correct first test. Only a model with stochastic retrieval practice learning predicted this effect. In conclusion, recall accuracy and recall latency suggest asymmetric learning, particularly in light of retrieval practice effects.

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106. Semisupervised Ordinal Regression Based on Empirical Risk Minimization

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Ordinal regression is aimed at predicting an ordinal class label. In this letter, we consider its semisupervised formulation, in which we have unlabeled data along with ordinal-labeled data to train an ordinal regressor. There are several metrics to evaluate the performance of ordinal regression, such as the mean absolute error, mean zero-one error, and mean squared error. However, the existing studies do not take the evaluation metric into account, restrict model choice, and have no theoretical guarantee. To overcome these problems, we propose a novel generic framework for semisupervised ordinal regression based on the empirical risk minimization principle that is applicable to optimizing all of the metrics mentioned above. In addition, our framework has flexible choices of models, surrogate losses, and optimization algorithms without the common geometric assumption on unlabeled data such as the cluster assumption or manifold assumption. We provide an estimation error bound to show that our risk estimator is consistent. Finally, we conduct experiments to show the usefulness of our framework.

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107. Bayesian Quadrature Optimization for Probability Threshold Robustness Measure

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In many product development problems, the performance of the product is governed by two types of parameters: design parameters and environmental parameters. While the former is fully controllable, the latter varies depending on the environment in which the product is used. The challenge of such a problem is to find the design parameter that maximizes the probability that the performance of the product will meet the

desired requisite level given the variation of the environmental parameter. In this letter, we formulate this practical problem as active learning (AL) problems and propose efficient algorithms with theoretically guaranteed performance. Our basic idea is to use a gaussian process (GP) model as the surrogate model of the product development process and then to formulate our AL problems as Bayesian quadrature optimization problems for probabilistic threshold robustness (PTR) measure. We derive credible intervals for the PTR measure and propose AL algorithms for the optimization and level set estimation of the PTR measure. We clarify the theoretical properties of the proposed algorithms and demonstrate their efficiency in both synthetic and real-world product development problems.